

Asymptotics for likelihood ratio tests of boundary points with singular information and unidentifiable nuisance parameters

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Abstract

We establish the asymptotic distribution of likelihood ratio tests (LRTs) in settings where some of the nuisance parameters are unidentifiable under the null hypothesis, parameters of interest lie on the boundary of the parameter space, and the information matrix of the identifiable parameters may be singular. Our work is motivated by mixture models and genetic linkage analysis, which exhibit all three features simultaneously, but it is applicable more broadly to other problems such as change-point detection. Under suitable regularity conditions, the asymptotic distribution of the LRT statistic under the null hypothesis is the supremum of a $\bar{\chi}^2$ -process, that is, a stochastic process whose marginal distributions are mixtures of χ^2 -distributions with weights depending on the nuisance parameter. Under local alternatives, the asymptotic distribution of the LRT statistic is the supremum of a noncentral $\bar{\chi}^2$ -process, whose marginal distributions are mixtures of truncated, noncentral χ^2 -distributions. In contrast to prior work on singular information, where singularity stems from the parameter of interest and changes the form of the limit distribution, here singularity is determined by the nuisance parameter and the limit has the same form as in the nonsingular case. Existing results for boundary inference with nonsingular information or without nuisance parameters are obtained as

special cases, and several existing application-specific results for mixture models and genetic linkage analysis are recovered and extended.

1 Introduction

Under regularity conditions, the likelihood ratio test (LRT) statistic has an asymptotic χ_p^2 -distribution under the null hypothesis, with the degrees of freedom p determined by the number of restrictions imposed by the null [Wilks, 1938]. In many applications, however, such regularity conditions do not hold and, consequently, inference based on the χ_p^2 -distribution is unreliable. We consider settings in which three nonregularities can be present simultaneously: (i) some nuisance parameters are unidentifiable under the null hypothesis; (ii) parameters of interest lie on the boundary of the parameter space under the null; and (iii) the information matrix for the parameters of interest may be singular at some values of the nuisance parameter. As we discuss in more detail shortly, these issues have been studied separately in the literature and in particular models, but not jointly in a general parametric or semi-parametric framework.

Important examples include LRTs for mixture models [Dacunha-Castelle and Gassiat, 1999, Chen and Chen, 2001, 2003] and genetic linkage analysis [Sham, 1998]. In mixture models, parameters of a component distribution are unidentifiable under a null hypothesis assigning zero weight to that component. Moreover, the mixing distribution may only be known to belong to an infinite-dimensional family. In genetic linkage analysis, the null hypothesis is that no disease gene is present along a chromosome of length T ; the alternative is that a gene is located at some position $\tau \in [0, T]$. The gene location τ is then unidentifiable under the null. The parameter of interest is the genetic variance attributable to the disease gene: it is nonnegative and equals zero under the null, placing it on the boundary of the parameter space, and the Fisher information can be singular. Further details on these examples, including references to previous work, are in Sections 5.1 and 5.2. Other problems for which the LRT statistic exhibits nonstandard asymptotics include testing the number of components in segmented regression [Feder, 1975, Davies, 1987], the closely related problem of testing whether a change point exists [Andrews and Ploberger, 1995], testing for a threshold effect in autoregression [Andrews and Ploberger, 1995], testing the relevance of some explanatory variables in nonlinear regression [Andrews and Ploberger, 1995], detection of a discrete frequency component of time series [Davies, 1987], and testing the order of an ARMA time series [Dacunha-Castelle and Gassiat, 1999].

Our main results give conditions under which the likelihood ratio statistic converges in law

to the supremum of a stochastic process, and we characterize that limiting process. Under the null hypothesis and appropriate regularity conditions, this process is a $\bar{\chi}^2$ -process, with marginal distributions that are mixtures of χ^2 -distributions and weights determined by the local geometry of the parameter space. Under contiguous (local) alternatives, the process is a noncentral $\bar{\chi}^2$ -process, with marginal distributions that are mixtures of truncated noncentral χ^2 -distributions, with weights not only determined by the local geometry of the parameter space, but also by the sequence of local alternatives.

The closest existing results are those of Andrews [2001], which concern boundary inference with nonsingular information and nulls that fix a parameter subvector to a point; we allow both singular information and more general composite null hypotheses. Even without these generalizations, however, the settings are complementary rather than nested. Our findings also recover, as special cases, results obtained by removing some of the complicating features. This includes classical results for boundary inference without nuisance parameters [Chernoff, 1954, Self and Liang, 1987, Geyer, 1994]. In addition, our framework incorporates results for testing whether a single parameter of interest is zero or positive, in the presence of nuisance parameters, when either all [Davies, 1977, 1987] or some [Ritz and Skovgaard, 2005] of these nuisance parameters are unidentifiable under the null hypothesis.

The source of singularity here differs from that in Rotnitzky et al. [2000], Ekvall and Bottai [2022], Bottai [2003], and Guédon et al. [2024]. There, the rank of the information matrix is determined by the parameter of interest, whereas here it is determined by the nuisance parameter. In Rotnitzky et al. [2000], singularity fundamentally changes the form of the asymptotic distribution, the likelihood is bimodal near the singular point, and only information matrices of rank one less than full are treated. Here, the rank at singular points is arbitrary and, under smoothness assumptions, the limit distribution has the same form as in the nonsingular case; the singularity complicates the analysis but not the conclusion.

2 Quadratic Approximation and $\bar{\chi}^2$ Process

We assume an underlying probability space $(\Omega, \mathcal{S}, \mathbf{P})$ and call any subset of Ω an event, whether it is measurable or not. Similarly, we call any function mapping elements of Ω to some set a random element of that set. In particular, let ℓ_n be a random real-valued function with domain Θ ; that is, for every $\omega \in \Omega$, $\ell_n(\cdot) = \ell_n(\cdot; \omega)$ is a real-valued function on Θ (the parameter space). Assume $\theta \in \Theta$ satisfies $\theta = (\xi, t)$ for t in some arbitrary set $\mathcal{T}$ and $\xi \in \Xi(t) \subseteq \mathbb{R}^p$. For now ℓ_n can be an arbitrary objective function, but later we assume it is a

log-likelihood, that ξ is a parameter of interest, and that t is a nuisance parameter.

A key object in our theory is the stochastic process X_n indexed by $\mathcal{T}$ and defined by

$$X_n(t) = \sup_{\xi \in \Xi(t)} 2\{\ell_n(\xi, t) - \ell_n(\xi_0, t)\}, \quad (1)$$

where ξ_0 is a particular value of ξ . Later, ξ_0 will be the null value of ξ . This corresponds to hypothesis testing with $H_0 : \xi = \xi_0$ versus $H_1 : \xi \in \Xi \setminus \{\xi_0\}$, where $\Xi = \cup_{t \in \mathcal{T}} \Xi(t)$. We will also assume that the nuisance parameter t is unidentifiable under the null, meaning that $\ell_n(\xi_0, t)$ does not depend on t . Then, the likelihood ratio test statistic for testing H_0 against H_1 is

$$\Lambda_n = \sup_{t \in \mathcal{T}} X_n(t). \quad (2)$$

First, however, we discuss the process X_n when ℓ_n is not necessarily the log-likelihood and Λ_n not necessarily a LRT statistic. For each $t \in \mathcal{T}$ and $n \in \{1, 2, \dots\}$, let $U_n(t)$ denote a random vector and let $I(t) \in \mathbb{R}^{p \times p}$ be deterministic and positive definite. Consider a quadratic expansion

$$\ell_n(\xi, t) - \ell_n(\xi_0, t) = \sqrt{n}(\xi - \xi_0)^T U_n(t) - \frac{n}{2}(\xi - \xi_0)^T I(t)(\xi - \xi_0) + n\|I(t)^{1/2}(\xi - \xi_0)\|_2^2 R_n(\xi, t), \quad (3)$$

where T denotes transpose, $\|\xi\|_2^2 = \xi^T \xi$ is the squared Euclidean norm in $\mathbb{R}^p$, and $R_n(\xi, t)$ is a remainder term to be specified. We call U_n and I a generalized score and information matrix, respectively. When ℓ_n is a log-likelihood, they can often be the usual score and Fisher information matrix for ξ , evaluated at ξ_0 . In that case, for a fixed t , (3) is the usual quadratic expansion used to derive the asymptotic distribution of the likelihood ratio test [see, for example, Serfling, 2002]. Here, by contrast, it will be important to control the behavior of U_n , I , and R_n as elements of appropriate spaces of functions on $\mathcal{T}$. Moreover, we will later consider cases where $I(t)$ is singular for some t .

Put $\delta = n^{1/2}A(t)^T(\xi - \xi_0)$ and $\tilde{\ell}_n(\delta, t) = \ell_n(\xi_0 + n^{-1/2}A(t)^{-T}\delta, t)$, where $A(t)A(t)^T = I(t)$. For example, $A(t)$ can be a Cholesky root or the symmetric matrix with the same eigenvectors as $I(t)$ and square-root eigenvalues. It follows that the set of possible values of δ is $\Delta_n(t) = n^{1/2}A(t)^T(\Xi(t) - \xi_0)$. Let also $Z_n(t) = A(t)^{-1}U_n(t)$ and $Q_n(\delta, t) = 2\delta^T Z_n(t) - \|\delta\|_2^2$. Then, by

(3), the function $\tilde{\ell}_n$ defined by $\tilde{\ell}_n(\delta, t) = \ell_n(\xi_0 + n^{-1/2}A(t)^{-\top}\delta, t)$ satisfies

$$2\{\tilde{\ell}_n(\delta, t) - \tilde{\ell}_n(0, t)\} = Q_n(\delta, t) + 2\|\delta\|_2^2 \tilde{R}_n(\delta, t), \quad (4)$$

where $\tilde{R}_n(\delta, t) = R_n(\xi_0 + n^{-1/2}A(t)^{-\top}\delta, t)$. Thus,

$$X_n(t) = \sup_{\delta \in \Delta_n(t)} \left\{ Q_n(\delta, t) + 2\|\delta\|_2^2 \tilde{R}_n(\delta, t) \right\}. \quad (5)$$

Under conditions in the next section, $Z_n \xrightarrow{\mathcal{L}} Z$ and $\Delta_n(t) \rightarrow \Delta(t)$, from which $X_n \xrightarrow{\mathcal{L}} X$ follows. The limit set $\Delta(t)$ encodes the boundary constraints on ξ : if ξ_0 is in the interior of $\Xi(t)$, then $\Delta(t) = \mathbb{R}^p$, whereas if ξ_0 lies on the boundary, $\Delta(t)$ is typically a proper convex cone. For each t , the limit process satisfies

$$X(t) = \sup_{\delta \in \Delta(t)} Q(\delta, t); \quad Q(\delta, t) = 2\delta^\top Z(t) - \|\delta\|_2^2. \quad (6)$$

Intuitively, then, the distribution of Λ_n will be close to that of

$$\Lambda = \sup_{t \in \mathcal{T}} X(t) \quad (7)$$

if, as functions on $\mathcal{T}$, Z_n , Δ_n , and $\tilde{R}_n(\delta, \cdot)$ are close to Z , Δ , and 0, respectively. Under certain conditions, we have

$$X(t) = \|Z(t)\|_2^2 - \inf_{\delta \in \Delta(t)} \|Z(t) - \delta\|_2^2 = \|Z(t)\|_2^2 - \|Z(t) - P_{\Delta(t)}Z(t)\|_2^2 = \|P_{\Delta(t)}Z(t)\|_2^2. \quad (8)$$

Specifically, the first equality is always true, while the second holds if the infimum is attained by some $\delta \in \Delta(t)$, in which case it defines the projection $P_{\Delta(t)}$. If $\Delta(t)$ is closed, the projection exists and is unique Lebesgue-almost everywhere [Geyer, 1994, Proposition 4.2]. The third equality holds if $\Delta(t)$ is a closed, convex cone.

Suppose in addition that Z is a centered Gaussian process whose marginal covariance matrix is the identity matrix Id_p of order p for all $t \in \mathcal{T}$. This will typically be the case when $\ell_n(\xi, t)$ is a log-likelihood and the null hypothesis $\xi = \xi_0$ holds. Then, for each fixed t , $X(t)$ has a χ^2 -distribution [Shapiro, 1985, 1988]. That is, for some nonnegative weights

$w_0(t), \dots, w_p(t)$ summing to one (whose values depend on $\Delta(t)$)

$$X(t) \sim \sum_{j=0}^p w_j(t) \chi_j^2. \quad (9)$$

Then, we say X is a $\bar{\chi}^2$ -process indexed by $\mathcal{T}$. Now suppose $\ell_n(\xi, t)$ is a log-likelihood and consider a sequence of contiguous (or local) alternatives $\theta_n = (\xi_n, t_0)$ with $\xi_n \rightarrow \xi_0$ at an appropriate rate. Then X_n converges weakly to a process X whose marginal distribution $X(t)$ is a mixture of truncated, noncentral χ_j^2 -distributions, and we refer to X as a noncentral $\bar{\chi}^2$ -process indexed by $\mathcal{T}$. The challenge here is not merely to show $X_n(t) \xrightarrow{\mathcal{L}} X(t)$ for fixed $t \in \mathcal{T}$, but rather to prove weak convergence $X_n \xrightarrow{\mathcal{L}} X$ on an appropriate function space.

3 Asymptotic Distributions

3.1 Asymptotics with uniformly positive definite information

We first consider the case where $I(t)$ is uniformly positive definite over $\mathcal{T}$, meaning its smallest eigenvalue is bounded away from zero uniformly in t . This simplifies the discussion and forms the basis for Section 3.3, where $I(t)$ may be singular.

Let $F^p(\mathcal{T})$ be the set of bounded $\mathbb{R}^p$ -valued functions on $\mathcal{T}$ (with $F^1 = F$); that is, $z \in F^p(\mathcal{T})$ if and only if $\|z\|_{\mathcal{T}} = \sup_{t \in \mathcal{T}} \|z(t)\|_2 < \infty$, where $\|\cdot\|_2$ is the Euclidean norm and $\|\cdot\|_{\mathcal{T}}$ is defined by the first equality. We equip $F^p(\mathcal{T})$ with the corresponding metric and Borel σ -algebra. Because some random elements we shall consider need not be measurable, some results concern outer probabilities and expectations, denoted by $\mathbf{P}^*$ and $\mathbf{E}^*$, respectively; we refer the reader to, for example, van der Vaart and Wellner [2023] for definitions and properties. We denote generic constants by c_i, C_i , with c_i small and C_i large, $i \in \{1, 2, \dots\}$.

Assumption 1. For every n , X_n is a random element of $F(\mathcal{T})$.

Assumption 2. Equation (3) holds with a remainder term that is stochastically, uniformly equicontinuous at ξ_0 in the sense that, for every $c_1, c_2 > 0$, there exists a $c_3 > 0$ such that

$$\limsup_{n \rightarrow \infty} \mathbf{P}^* \left(\sup_{t \in \mathcal{T}} \sup_{\xi \in \bar{B}_{c_3}(\xi_0; t)} |R_n(\xi, t)| > c_1 \right) < c_2, \quad (10)$$

where $\bar{B}_{c_3}(\xi_0; t) = \{\xi \in \Xi(t) : \|\xi - \xi_0\|_2 \leq c_3\}$.

Assumption 3. Approximate partial maximizers are uniformly consistent; that is, if for every n and $t \in \mathcal{T}$, $\hat{\xi}_n(t)$ satisfies $\ell_n(\hat{\xi}_n(t), t) \geq \sup_{\xi \in \Xi(t)} \ell_n(\xi, t) - O_{p^*}(1)$, with the $O_{p^*}(1)$ term not depending on t , then $\|\hat{\xi}_n - \xi_0\|_{\mathcal{T}} = o_{p^*}(1)$.

Assumption 4. For some constants $\underline{\kappa}$ and $\bar{\kappa}$,

$$0 < \underline{\kappa} \leq \inf_{t \in \mathcal{T}} \lambda_{\min}\{I(t)\} \leq \sup_{t \in \mathcal{T}} \lambda_{\max}\{I(t)\} \leq \bar{\kappa} < \infty,$$

where $\lambda_{\min}(\cdot)$ and $\lambda_{\max}(\cdot)$ are, respectively, the smallest and largest eigenvalues of the argument matrix.

Assumption 5. There is a tight, Borel-measurable $Z \in F^p(\mathcal{T})$ such that $Z_n \xrightarrow{\mathcal{L}} Z$ in $F^p(\mathcal{T})$ as $n \rightarrow \infty$.

Assumption 6. For every $t \in \mathcal{T}$, $\Xi(t)$ is closed.

Assumption 7. For every $t \in \mathcal{T}$, there is a closed cone $C(t) \subseteq \mathbb{R}^p$ such that

$$\limsup_{s \downarrow 0} \sup_{t \in \mathcal{T}} d[s^{-1}\{\Xi(t) - \xi_0\}, C(t)] = 0, \quad (11)$$

where $d(\cdot, \cdot)$ is the distance for sets defined in Section A.1 of the Supplementary material.

Assumption 1 ensures X_n and its limit X are elements of the same metric space. It can be relaxed, for instance by working with a truncated version $\min(X_n, n) \in F(\mathcal{T})$; assuming $\Lambda_n = O_{p^*}(1)$, we have $\mathbf{P}^*(\Lambda_n \leq n) \rightarrow 1$ and hence the truncation has no effect on the asymptotic distribution. Neither X_n nor Z_n need be measurable.

In settings with independent and identically distributed (i.i.d.) observations, Assumption 2 can often be verified with U_n the score function and I the Fisher information matrix for one observation. This is formalized in Proposition 4.1.

If $\mathcal{T}$ is a singleton, which essentially corresponds to a setting without nuisance parameters, Assumptions 2–7 are similar to common ones [see, for example, Geyer, 1994]. We are mostly interested in cases where $\mathcal{T}$ is a more complicated set. The results in this section make few explicit assumptions about $\mathcal{T}$, though verifying Assumptions 2–7 typically requires some structure. For example, in some settings $I(\cdot)$ is a continuous function from $\mathbb{R}^q$ to $\mathbb{R}^{p \times p}$, and then Assumption 4 may require $\mathcal{T}$ to be a compact subset of $\mathbb{R}^q$. Similarly, verifying Assumption 5 often amounts to showing a class of functions indexed by $\mathcal{T}$ is Donsker, which in general requires controlling the complexity of that class [see, for example, van der Vaart

and Wellner, 2023, Section 2]. More generally, Z need not be a Gaussian process, but it will typically be so when ℓ_n is a log-likelihood. In settings with i.i.d. observations, for example, Z is often a Gaussian process, which is centered under the null hypothesis $H_0 : \xi = \xi_0$ and non-centered under contiguous alternatives (see Section 4).

Assumption 3 typically requires case-by-case verification, as sufficient conditions depend on the specific model and parameter space. Assumption 4 is common in the literature but will be relaxed in Section 3.3, at the expense of making stronger assumptions about $\mathcal{T}$, among other things. Assumption 6 ensures $\Delta_n(t)$ is closed, so that projections onto $\Delta_n(t)$ and $\Delta(t)$ exist. It can be relaxed if this can be ensured by other means. Assumption 7 is essentially a uniform version of Chernoff regularity [Chernoff, 1954], which is common in the literature. It can only hold if $\xi_0 \in \Xi(t)$ for all $t \in \mathcal{T}$. With the $C(t)$ given by Assumption 7, let $\Delta(t) = A(t)^T C(t)$, as in the definition of $X(t)$ in (6).

We are ready to state the main result of the section, along with two illustrative corollaries. The proof invokes several lemmas, which we state and discuss shortly. With the lemmas in hand, the proof is a straightforward application of a continuous mapping theorem and Slutsky's lemma. Therefore, it is deferred to the Supplementary material, along with the proofs of other results stated without proof in the main text.

Theorem 3.1. *Under Assumptions 1-7, X is a Borel-measurable random element of $F(\mathcal{T})$ and $X_n \xrightarrow{\mathcal{L}} X$ in $F(\mathcal{T})$ as $n \rightarrow \infty$.*

Corollary 3.2. *Under Assumptions 1-7, and assuming $t \mapsto \ell_n(\xi_0, t)$ is constant, $\Lambda = \|X\|_{\mathcal{T}}$ in (7) is Borel-measurable. Moreover, for $\Lambda_n = \|X_n\|_{\mathcal{T}}$ in (2), $\Lambda_n \xrightarrow{\mathcal{L}} \Lambda$ as $n \rightarrow \infty$.*

Corollary 3.2 is an immediate consequence of Theorem 3.1 and the Continuous Mapping Theorem. Unless we also assume $C(t)$ is convex for every $t \in \mathcal{T}$, X need not be a $\bar{\chi}^2$ -process, as discussed following (6).

Remark. Our principal application of Theorem 3.1 and Corollary 3.2 is when $\ell_n(\xi, t)$ is a log-likelihood and $\ell_n(\xi_0, t)$ does not depend on $t \in \mathcal{T}$. Then Λ_n is the likelihood ratio test (LRT) statistic for testing the simple null hypothesis $H_0 : \xi = \xi_0$ against $H_1 : \xi \in \Xi \setminus \{\xi_0\}$. If $\xi = \xi_0$, the convergences in Theorem 3.1 and Corollary 3.2 provide weak convergence of X_n and the LRT statistic Λ_n under H_0 , respectively. In particular, the significance level $\alpha_n(c) = \mathbb{P}_{(\xi_0, t)}(\Lambda_n \geq c)$ of the LRT with threshold $c > 0$ is independent of t and

$$\alpha_n(c) \rightarrow \alpha(c) = \mathbb{P}_{(\xi_0, t)}(\Lambda \geq c) \text{ as } n \rightarrow \infty \quad (12)$$

at all continuity points c of the distribution function F_Λ of Λ under H_0 . Theorem 3.1 can also be applied to weak convergence of X_n under contiguous alternatives $\theta_n = (\xi_n, t_0)$, if $\xi_n \rightarrow \xi_0$ at an appropriate rate. Then $\alpha_n(c; \xi_n, t_0) = P_{(\xi_n, t_0)}(\Lambda_n \geq c)$ converges to a limiting power function. It is also possible to consider limiting averaged local power functions [Andrews and Ploberger, 1995, Andrews, 1996], where a limiting averaged power of the LRT is evaluated over a set of local alternatives that correspond to the same asymptotic noncentrality parameter. $\square$

The following corollary shows how classical results for boundary points are obtained as special cases by taking, for example, $\mathcal{T} = \{0\}$; the choice is arbitrary as the value of t does not affect the result.

Corollary 3.3. *Under the conditions of Corollary 3.2, suppose there are no nuisance parameters, i.e. $\mathcal{T} = \{0\}$, and additionally that $\Delta = \Delta(0)$ is a closed convex cone and $Z(0) \sim N(\mu, Id_p)$ for some $\mu \in \mathbb{R}^p$. If $\mu = 0$, then $\Lambda_n \xrightarrow{\mathcal{L}} \Lambda = X(0) = \|P_\Delta Z(0)\|_2^2 \sim \bar{\chi}_p^2$ as $n \rightarrow \infty$, where the limiting $\bar{\chi}_p^2$ -distribution in (9) has weights $w_0, w_1, \dots, w_p$ determined by Δ . If additionally ξ_0 is in the interior of Ξ , then $\Delta = \mathbb{R}^p$ and $X(0) \sim \chi_p^2$. If $\mu \neq 0$, then $\Lambda = \|P_\Delta[\mu + \varepsilon(0)]\|_2^2$, with $\varepsilon(0) \sim N(0, Id_p)$, will be a mixture of truncated, noncentral χ^2 -distributions, with weights depending on μ . If additionally ξ_0 is in the interior of Ξ , then $X(0) \sim \chi_{p, \|\mu\|_2^2}^2$ has a noncentral χ^2 -distribution with noncentrality parameter $\|\mu\|_2^2$.*

As we will find in Section 4 for i.i.d. data, the case $\mu = 0$ of Corollary 3.3 corresponds to asymptotic limits under the null hypothesis H_0 , whereas $\mu \neq 0$ corresponds to limits for a sequence of contiguous alternatives.

Now, to discuss the components of the proof of Theorem 3.1 define, for every $t \in \mathcal{T}$,

$$\tilde{X}_n(t) = \sup_{\delta \in \Delta_n(t)} Q_n(\delta, t); \quad \check{X}_n(t) = \sup_{\delta \in \Delta(t)} Q_n(\delta, t). \quad (13)$$

The proof strategy is to show that X_n is close to $\tilde{X}_n$, which in turn is close to $\check{X}_n$, which converges in law to X . We shall need the following rate of convergence result.

Lemma 3.4. *Under Assumptions 2–5, every $\hat{\xi}_n$ in Assumption 3 satisfies*

$$\|\hat{\xi}_n - \xi_0\|_{\mathcal{T}} = O_{P^*}(n^{-1/2}).$$

Lemma 3.5. *Under Assumptions 2–5, $\tilde{X}_n$ is a random element of $F(\mathcal{T})$ for all n , and $\|\tilde{X}_n - X_n\|_{\mathcal{T}} = o_{P^*}(1)$ as $n \rightarrow \infty$.*

Proof. We first show $\tilde{X}_n$ is a random element of $F(\mathcal{T})$ by an argument we will use variations on several times: since $0 \in \Delta_n(t)$ and $Q_n(0, t) = 0$, the supremum of $\delta \mapsto Q_n(\delta, t)$ over $\Delta_n(t)$ is non-negative. But for any δ with $\|\delta\|_2 > 2\|Z_n(t)\|_2$, by Cauchy–Schwarz,

$$Q_n(\delta, t) \leq 2\|\delta\|_2\|Z_n(t)\|_2 - \|\delta\|_2^2 = \|\delta\|_2(2\|Z_n(t)\|_2 - \|\delta\|_2) < 0. \quad (14)$$

Thus,

$$|\tilde{X}_n(t)| = \sup_{\delta \in \Delta_n(t), \|\delta\|_2 \leq 2\|Z_n(t)\|_2} Q_n(\delta, t) \leq 8\|Z_n(t)\|_2^2,$$

which, as a function of t , is a random element of $F(\mathcal{T})$ since Z_n is a random element of $F^p(\mathcal{T})$ by Assumption 5.

Next, pick $\hat{\xi}_n(t)$ such that $\ell_n(\hat{\xi}_n(t), t) \geq \sup_{\xi \in \Xi(t)} \ell_n(\xi, t) - 1/n$, say. Let $\hat{\delta}_n(t) = n^{1/2}A(t)(\hat{\xi}_n(t) - \xi_0)$ and note $\|\hat{\delta}_n\|_{\mathcal{T}} = O_{p^*}(1)$ by Lemma 3.4 and Assumption 4. Similarly, pick a $\tilde{\delta}_n(t)$ such that $Q_n(\tilde{\delta}_n(t), t) \geq \sup_{\delta \in \Delta_n(t)} Q_n(\delta, t) - 1/n$. That supremum is non-negative by the arguments preceding (14), so $\tilde{\delta}_n(t)$ must achieve at least the value $-1/n$. We will use this to show $\|\tilde{\delta}_n\|_{\mathcal{T}} = O_{p^*}(1)$. To that end, suppose $\|Z_n\|_{\mathcal{T}} < C_1$. Then by the inequalities in (14), any $\|\delta\|_2 > 3C_1$ leads to a value less than $-3C_1$, which is less than $-1/n$ for all large enough n . Thus, for such n , $\|\tilde{\delta}_n\|_{\mathcal{T}} \leq 3C_1$ whenever $\|Z_n\|_{\mathcal{T}} \leq C_1$, and since $\|Z_n\|_{\mathcal{T}} = O_{p^*}(1)$, we have $\|\tilde{\delta}_n\|_{\mathcal{T}} = O_{p^*}(1)$.

Next, for any $c_1, C_1 > 0$, by countable sub-additivity of outer probabilities, $\mathbf{P}^*(\|X_n - \tilde{X}_n\|_{\mathcal{T}} > c_1)$ is upper bounded by

$$\mathbf{P}^*(\|X_n - \tilde{X}_n\|_{\mathcal{T}} > c_1, \|\hat{\delta}_n\|_{\mathcal{T}} \leq C_1, \|\tilde{\delta}_n\|_{\mathcal{T}} \leq C_1) + \mathbf{P}^*(\|\hat{\delta}_n\|_{\mathcal{T}} > C_1) + \mathbf{P}^*(\|\tilde{\delta}_n\|_{\mathcal{T}} > C_1). \quad (15)$$

The last two terms can be made arbitrarily small by choosing C_1 large enough, so let us deal with the first term. Define

$$\Upsilon_{C_1, n} = \{(\delta, t) : t \in \mathcal{T}, \delta \in \Delta_n(t), \|\delta\|_2 \leq C_1\}.$$

Since $\hat{\delta}_n(t)$ and $\tilde{\delta}_n(t)$ are $1/n$ -approximate maximizers satisfying $\|\cdot\|_2 \leq C_1$, two applications of the approximate-maximizer property and $|\sup f - \sup g| \leq \sup |f - g|$ give

$$\|X_n - \tilde{X}_n\|_{\mathcal{T}} \leq 4/n + \sup_{(\delta, t) \in \Upsilon_{C_1, n}} \left| 2\{\tilde{\ell}_n(\delta, t) - \tilde{\ell}_n(0, t)\} - Q_n(\delta, t) \right|.$$

The quantity in absolute value in the last display is, by (4) and the definition of $\tilde{R}_n$,

$2\|\delta\|_2^2|R_n(\xi_0+n^{-1/2}A(t)^{-\top}\delta, t)|$. Since, for $(\delta, t) \in \Upsilon_{C_1, n}$, $|n^{-1/2}A(t)^{-\top}\delta| \leq n^{-1/2}\|A(t)^{-\top}\|\|\delta\|_2 \leq n^{-1/2}C_1\bar{\kappa}^{-1/2}$, $\xi_0 + n^{-1/2}A(t)^{-\top}\delta \rightarrow \xi_0$, (10) gives that the last supremum in the last display is $o_{p^*}(1)$, which finishes the proof. $\square$

Lemma 3.6. *Under Assumptions 2–7, $\tilde{X}_n$ and $\check{X}_n$ are random elements of $F(\mathcal{T})$ for all n , and $\|\tilde{X}_n - \check{X}_n\|_{\mathcal{T}} = o_{p^*}(1)$ as $n \rightarrow \infty$.*

Proof. That $\tilde{X}_n \in F(\mathcal{T})$ is already given by Lemma 3.5. That $\check{X}_n \in F(\mathcal{T})$ follows from an almost identical argument, and will be established in the proof of Lemma 3.7.

Since $\Xi(t)$ is closed by Assumption 6, so is $\Delta_n(t)$, and therefore the projection onto it exists. Similarly, by Assumption 7, the projection onto $\Delta(t)$ exists. Thus, as in (6), we get $\tilde{X}_n(t) = \|Z_n(t)\|_2^2 - \|Z_n(t) - P_{\Delta_n(t)}Z_n(t)\|_2^2$ and $\check{X}_n(t) = \|Z_n(t)\|_2^2 - \|Z_n(t) - P_{\Delta(t)}Z_n(t)\|_2^2$. Thus, for outcomes where $\|Z_n\|_{\mathcal{T}} \leq C_1$ we get that $\sup_{t \in \mathcal{T}} |\tilde{X}_n(t) - \check{X}_n(t)|$ equals

$$\sup_{t \in \mathcal{T}} \left| \|Z_n(t) - P_{\Delta(t)}Z_n(t)\|_2^2 - \|Z_n(t) - P_{\Delta_n(t)}Z_n(t)\|_2^2 \right| \leq \sup_{t \in \mathcal{T}, \|z\|_2 \leq C_1} \left| \|z - P_{\Delta(t)}z\|_2^2 - \|z - P_{\Delta_n(t)}z\|_2^2 \right|,$$

which tends to zero by Lemma A.1. Since $\|Z_n\|_{\mathcal{T}} \xrightarrow{\mathcal{L}} \|Z\|_{\mathcal{T}}$ by continuous mapping theorem and the limit is tight by Assumption 5, $\|Z_n\|_{\mathcal{T}}$ is asymptotically tight. Thus, the outer probability of $\|Z_n\|_{\mathcal{T}} > C_1$ can be made arbitrarily small asymptotically by choosing C_1 large enough, which completes the proof. $\square$

Lemma 3.7. *Under Assumptions 4–7, $\check{X}_n$ is a random element of $F(\mathcal{T})$ for all n , X is a Borel-measurable random element of $F(\mathcal{T})$, and $\check{X}_n \xrightarrow{\mathcal{L}} X$ in $F(\mathcal{T})$ as $n \rightarrow \infty$.*

Proof. Consider the mapping $g : F^p(\mathcal{T}) \rightarrow F(\mathcal{T})$ defined for $z \in F^p(\mathcal{T})$ by $g(z)(t) = \sup_{\delta \in \Delta(t)} (2\delta^\top z(t) - \|\delta\|_2^2)$. Indeed, $g(z) \in F(\mathcal{T})$ for any $z \in F^p(\mathcal{T})$ since, by the triangle inequality and $0 \in \Delta(t)$, arguing as in (6),

$$|g(z)(t)| \leq \sup_{t \in \mathcal{T}} \|z(t)\|_2^2 + \sup_{t \in \mathcal{T}} \inf_{\delta \in \Delta(t)} \|z(t) - \delta\|_2^2 \leq 2\|z\|_{\mathcal{T}}^2 < \infty.$$

This also shows that $X = g(Z)$ and $\check{X}_n = g(Z_n)$ are random elements of $F(\mathcal{T})$ since Z and Z_n are random elements of $F^p(\mathcal{T})$ by Assumption 5.

To show g is continuous, pick an arbitrary $z_\infty \in F^p(\mathcal{T})$ and $z_m \in F^p(\mathcal{T})$ such that $\|z_\infty - z_m\|_{\mathcal{T}} \rightarrow 0$ as $m \rightarrow \infty$. Pick a finite $C_1 > \|z_\infty\|_{\mathcal{T}}$ so that $\|z_m\|_{\mathcal{T}} < C_1$ for all large enough m . Then, as argued in the proof of Lemma 3.5, any δ with $\|\delta\|_2 > 3C_1$ leads to

$\delta^\top z_m(t) - \|\delta\|_2^2 < 0$, which since $0 \in \Delta(t)$ implies

$$g(z_m)(t) = \sup_{\delta \in \Delta(t) \cap \bar{B}_{3C_1}} (2\delta^\top z_m(t) - \|\delta\|_2^2),$$

also when $m = \infty$. Thus, using that the difference of suprema is no greater than the supremum of the difference, we get

$$|g(z_m)(t) - g(z_\infty)(t)| \leq \sup_{\delta \in \Delta(t) \cap \bar{B}_{3C_1}} |2\delta^\top (z_m(t) - z_\infty(t))| \leq 6C_1 \|z_m(t) - z_\infty(t)\|_2.$$

Taking the supremum over $t \in \mathcal{T}$ on both sides and sending $m \rightarrow \infty$ shows $\|g(z_m) - g(z_\infty)\|_{\mathcal{T}} \rightarrow 0$, so g is continuous. Thus, X is Borel-measurable since Z is, and since $Z_n \xrightarrow{\mathcal{L}} Z$ by Assumption 5, the continuous mapping theorem gives $\check{X}_n = g(Z_n) \xrightarrow{\mathcal{L}} g(Z) = X$, which completes the proof. $\square$

With these lemmas in hand, the proof of Theorem 3.1 is a straightforward application of Slutsky's lemma and the continuous mapping theorem given in the Supplementary material.

3.2 Likelihood ratio test with composite null

In this section we will generalize to a composite null hypothesis, and the corresponding alternative hypothesis, of the form

$$\begin{aligned} H_0 : \quad & \xi \in \Xi_0, \\ H_1 : \quad & \xi \in \Xi \setminus \Xi_0, \end{aligned} \tag{16}$$

where $\Xi_0 \subseteq \mathbb{R}^p$ is such that, for each $\xi \in \Xi_0$, there exists a $t \in \mathcal{T}$ with $(\xi, t) \in \Theta$. For each $t \in \mathcal{T}$, let $\Xi^0(t)$ be the set of $\xi \in \Xi_0$ such that $(\xi, t) \in \Theta$. Thus, $\Xi_0 = \cup_{t \in \mathcal{T}} \Xi^0(t)$, and when ℓ_n is a log-likelihood the likelihood ratio test statistic for testing H_0 against H_1 in (16) is

$$\Lambda_n = 2 \left\{ \sup_{t \in \mathcal{T}} \sup_{\xi \in \Xi(t)} \ell_n(\xi, t) - \sup_{t \in \mathcal{T}} \sup_{\xi \in \Xi^0(t)} \ell_n(\xi, t) \right\}. \tag{17}$$

To obtain the asymptotic distribution for this case, define X_n^0 like X_n but with $\Xi^0(t)$ in place of $\Xi(t)$; that is,

$$X_n^0(t) = 2 \sup_{\xi \in \Xi^0(t)} \{\ell_n(\xi, t) - \ell_n(\xi_0, t)\}. \tag{18}$$

The joint convergence of (X_n, X_n^0) , given by the following result, will lead to the asymptotic distribution of Λ_n under the null. Assumption 7 can only apply to $\Xi^0(t)$ if $\xi_0 \in \Xi^0(t)$ for all $t \in \mathcal{T}$. In a hypothesis testing context, this means the null hypothesis must be correct regardless of the value of the nuisance parameter.

Lemma 3.8. *Under Assumptions 1–7, with Assumptions 6 and 7 also holding for $\Xi^0(t)$ and a cone $C^0(t)$ in place of $\Xi(t)$ and $C(t)$, respectively, $(X_n, X_n^0) \xrightarrow{\mathcal{L}} (X, X^0)$ in $F^2(\mathcal{T})$. Here $X^0 = \{X^0(t); t \in \mathcal{T}\}$, where $X^0(t)$ is defined as in (6), with $\Delta^0(t) = A(t)^\top C^0(t)$ instead of $\Delta(t) = A(t)^\top C(t)$.*

Proof. The proof follows the same approximation and continuous mapping steps as for Theorem 3.1. Under the assumptions of Theorem 3.1, and the corresponding assumptions with $\Xi^0(t)$ in place of $\Xi(t)$, the lemmas leading up to Theorem 3.1 yield $\|X_n - \check{X}_n\|_{\mathcal{T}} = o_{\mathbb{P}^*}(1)$ and $\|X_n^0 - \check{X}_n^0\|_{\mathcal{T}} = o_{\mathbb{P}^*}(1)$, where $\check{X}_n^0(t) = \sup_{\delta \in \Delta^0(t)} (2\delta^\top Z_n(t) - \|\delta\|_2^2)$. For $z \in F^p(\mathcal{T})$, define maps $g, g_0 : F^p(\mathcal{T}) \rightarrow F(\mathcal{T})$ by

$$g(z)(t) = \sup_{\delta \in \Delta(t)} (2\delta^\top z(t) - \|\delta\|_2^2), \quad g_0(z)(t) = \sup_{\delta \in \Delta^0(t)} (2\delta^\top z(t) - \|\delta\|_2^2).$$

Then $\check{X}_n = g(Z_n)$ and $\check{X}_n^0 = g_0(Z_n)$. By the argument in Lemma 3.7, applied to Δ and, separately, to Δ^0 , both g and g_0 are continuous. Hence, the map $z \mapsto (g(z), g_0(z))$ from $F^p(\mathcal{T})$ to $F^2(\mathcal{T})$ is continuous, and since $Z_n \xrightarrow{\mathcal{L}} Z$ in $F^p(\mathcal{T})$ by Assumption 5, the continuous mapping theorem gives $(\check{X}_n, \check{X}_n^0) = (g(Z_n), g_0(Z_n)) \xrightarrow{\mathcal{L}} (g(Z), g_0(Z)) = (X, X^0)$ in $F^2(\mathcal{T})$. The $o_{\mathbb{P}^*}(1)$ approximations then imply $(X_n, X_n^0) \xrightarrow{\mathcal{L}} (X, X^0)$ by Slutsky's lemma. $\square$

Theorem 3.9. *Under the conditions of Lemma 3.8, and assuming $t \mapsto \Xi^0(t)$ and $t \mapsto \ell_n(\xi, t)$, for all $\xi \in \Xi_0$, are constant, the random variable defined in (17) converges weakly $\Lambda_n \xrightarrow{\mathcal{L}} \Lambda = \|X - X^0\|_{\mathcal{T}}$ as $n \rightarrow \infty$.*

Proof. Since $\Xi^0(t)$ does not depend on t , it is equal to Ξ_0 . Since also $\ell_n(\xi, t)$ is constant in t for $\xi \in \Xi_0$, $X_n^0(t) = 2\{\sup_{\xi \in \Xi_0} \ell_n(\xi, t) - \ell_n(\xi_0, t)\}$ in fact does not depend on t . Thus, since $\xi_0 \in \Xi_0$ by Assumption 7, we can move the supremum over t outside the brackets in (17), add and subtract $\ell_n(\xi_0)$, and get

$$\Lambda_n = \sup_{t \in \mathcal{T}} \{X_n(t) - X_n^0(t)\} = \|X_n - X_n^0\|_{\mathcal{T}} = \|X_n\|_{\mathcal{T}} - \|X_n^0\|_{\mathcal{T}}.$$

By Lemma 3.8 and the continuous mapping theorem, $\|X_n\|_{\mathcal{T}} - \|X_n^0\|_{\mathcal{T}}$ converges in law to $\|X\|_{\mathcal{T}} - \|X^0\|_{\mathcal{T}} = \|X - X^0\|_{\mathcal{T}}$. $\square$

An interesting special case of the setting in Theorem 3.9 is when $C^0 = \lim_{s \downarrow 0} s^{-1}(\Xi_0 - \xi_0)$ is a linear subspace. This happens, for example, when some components of ξ are restricted by the null hypothesis, but other components are free. For instance, if $\xi = (\xi^{(1)}, \xi^{(2)}) \in \Xi(t) = \Xi = [0, \infty)^{p_1} \times \mathbb{R}^{p_2}$, and $\Xi_0 = \{\xi \in \Xi : \xi^{(1)} = \xi_0^{(1)}\}$, then $C^0 = \{0\} \times \mathbb{R}^{p_2}$. In such settings, the asymptotic distribution is similar to when testing a simple null of dimension p_1 , as formalized by the following result; compare to Corollary 3.2.

Theorem 3.10. *Assume the conditions of Theorem 3.9 hold, and that for each $t \in \mathcal{T}$, $\Delta(t)$ is a closed convex cone and $\Delta^0(t) = A(t)^\top C^0$ is a linear subspace. Then $K(t) = \Delta(t) \cap \{\Delta^0(t)\}^\perp$ is a closed convex cone, $\Delta(t) = \Delta^0(t) \oplus K(t)$, and, consequently, $\Lambda = \sup_{t \in \mathcal{T}} \|P_{K(t)} Z(t)\|_2^2$.*

Since $K(t)$ is a closed convex cone, the same argument as in (6) shows that the distribution of $X(t) - X^0(t) = \|P_{K(t)} Z(t)\|_2^2$ is a mixture of χ^2 -distributions when $Z(t)$ is a centered Gaussian process, with weights determined by the geometry of $K(t)$. Then Λ is the supremum of a $\bar{\chi}^2$ -process. If $K(t)$ is polyhedral, defined by $r(t)$ linear inequality constraints, only the $\chi_{p_1-r(t)}^2, \dots, \chi_{p_1}^2$ components receive positive weights.

Remark. Suppose $\ell_n(\xi, t)$ is a log-likelihood and the conditions of Theorem 3.9 hold. If $\xi = \xi_0$ for some $\xi_0 \in \Xi_0$, Theorem 3.9 provides weak convergence of the likelihood ratio test (LRT) statistic Λ_n under an element ξ_0 of the composite null hypothesis H_0 in (16). If additionally the conditions of Theorem 3.10 hold, the distribution of the limiting LRT statistic Λ in Theorem 3.10 is typically independent of the choice of $\xi_0 \in \Xi_0$. If H_0 is rejected when $\Lambda_n \geq c$, the significance level satisfies

$$\alpha_n(c; \xi_0) = P_{(\xi_0, t)}(\Lambda_n \geq c) \rightarrow \alpha(c; \xi_0) = P_{(\xi_0, t)}(\Lambda \geq c) \quad (19)$$

as $n \rightarrow \infty$ at all continuity points c of the distribution function F_Λ of Λ , independently of $t \in \mathcal{T}$. Note that the limiting significance level $\alpha(c) = \alpha(c; \xi_0)$ typically does not depend on the choice of $\xi_0 \in \Xi_0$ whenever the conditions of Theorem 3.10 hold. Suppose instead we have a sequence of contiguous alternatives $\theta_n = (\xi_n, t_0)$ such that $\xi_n \rightarrow \xi_0$ as $n \rightarrow \infty$ for some $\xi_0 \in \Xi_0$. Then $\alpha_n(c; \xi_n, t_0) = P_{(\xi_n, t_0)}(\Lambda_n \geq c)$ corresponds to the power of the LRT and if $\xi_n \rightarrow \xi_0$ at an appropriate rate it will converge to a limiting power function. As mentioned in Section 3.1, it is also possible to consider limiting averaged local power functions [Andrews and Ploberger, 1995, Andrews, 1996]. $\square$

3.3 Singular information

In some cases the information matrix $I(t)$ is singular for some $t \in \mathcal{T}$. We partition $\mathcal{T}$ into $\mathcal{T}_s = \{t \in \mathcal{T} : \lambda_{\min}\{I(t)\} = 0\}$ and $\mathcal{T}_{ns} = \{t \in \mathcal{T} : \lambda_{\min}\{I(t)\} > 0\}$. We will first give conditions for convergence in law of $\Lambda_n^{(ns)} = \sup_{t \in \mathcal{T}_{ns}} X_n(t)$, and then extend that to $\Lambda_n = \sup_{t \in \mathcal{T}} X_n(t)$.

Establishing the asymptotic distribution of $\Lambda_n^{(ns)}$ is more complicated than applying the conditions from Section 3 with $\mathcal{T}$ replaced by $\mathcal{T}_{ns}$. In particular, assuming something like Assumption 4 on $\mathcal{T}_{ns}$ is inappropriate as, in many settings of interest, one can find a sequence (t_m) in $\mathcal{T}_{ns}$ tending to a $t_* \in \mathcal{T}_s$ and get $\lambda_{\min}\{I(t_m)\} \rightarrow 0$ as $m \rightarrow \infty$. Our strategy will be to first work on sets $\mathcal{T}_\epsilon = \{t \in \mathcal{T} : d(t, \mathcal{T}_s) \geq \epsilon\}$ for $\epsilon > 0$. This requires a metric d on $\mathcal{T}$ and, for sets $A \subseteq \mathcal{T}$, defining $d(t, A) = \inf_{t' \in A} d(t, t')$.

Assumption 8. The set $\mathcal{T}$ is a metric space with metric $d(\cdot, \cdot)$, $\mathcal{T}_s$ is closed in $\mathcal{T}$, and $\mathcal{T}_{ns}$ is dense in $\mathcal{T}$.

Closedness of $\mathcal{T}_s$ ensures $d(t, \mathcal{T}_s) > 0$ for every $t \in \mathcal{T}_{ns}$, so that $\cup_{\epsilon > 0} \mathcal{T}_\epsilon = \mathcal{T}_{ns}$. Density of $\mathcal{T}_{ns}$ is used when extending the result from $\mathcal{T}_{ns}$ to all of $\mathcal{T}$ in Corollary 3.12.

Assumption 2 does not control the remainder term $\tilde{R}_n(\delta, t)$ uniformly over $\mathcal{T}_{ns}$ unless the eigenvalues of $I(t)$ are bounded away from zero. Since this cannot be assumed on $\mathcal{T}_{ns}$, we instead make the following assumption (which, under Assumption 4 is equivalent to Assumption 2):

Assumption 2*. For every $t \in \mathcal{T}_{ns}$, (5) holds with a remainder term $\tilde{R}_n(\delta, t)$ that satisfies the following: for every $c_1, c_2 > 0$, there exists a $c_3 > 0$ such that

$$\limsup_{n \rightarrow \infty} \mathbf{P}^* \left(\sup_{t \in \mathcal{T}_{ns}} \sup_{\delta \in \tilde{B}(n, t, c_3)} \left| \tilde{R}_n(\delta, t) \right| > c_1 \right) < c_2,$$

where $\tilde{B}(n, t, c_3) = \{\delta \in \Delta_n(t) : \|\delta\|_2 \leq \sqrt{nc_3}\}$.

Assumption 4* modifies Assumption 4 by allowing $\underline{\kappa}$ to depend on ϵ . The lower bound on $\lambda_{\min}\{I(t)\}$, which is usually the challenging part, holds if $I(\cdot)$ is continuous on a compact $\mathcal{T}_\epsilon$, for example.

Assumption 4*. There exists a $\bar{\kappa} < \infty$ and, for every $\epsilon > 0$, a $\underline{\kappa}_\epsilon > 0$ such that, for every $t \in \mathcal{T}_\epsilon$,

$$\underline{\kappa}_\epsilon \leq \lambda_{\min}\{I(t)\} \leq \lambda_{\max}\{I(t)\} \leq \bar{\kappa}.$$

Assumption 5^{*} is Assumption 5 restricted to $\mathcal{T}_{ns}$. In particular, $Z_n(t) = A(t)^{-1}U_n(t)$ is only defined for $t \in \mathcal{T}_{ns}$. The spaces $F^p(\mathcal{T}_{ns})$, $p \geq 1$, are defined as in Section 3.1 but with $\mathcal{T}_{ns}$ in place of $\mathcal{T}$.

Assumption 5^{*}. There is a tight and Borel-measurable $Z \in F^p(\mathcal{T}_{ns})$ such that $Z_n \xrightarrow{\mathcal{L}} Z$ in $F^p(\mathcal{T}_{ns})$ as $n \rightarrow \infty$, with Z_n defined as in (4).

Assumption 9 ensures $\Delta_n(t) \subseteq \Delta(t)$ for all n , which gives $\tilde{X}_n(t) \leq \check{X}_n(t)$ and is key to the bounding argument in the proof. It is satisfied whenever $\Xi(t)$ is convex for every $t \in \mathcal{T}$.

Assumption 9. For every $t \in \mathcal{T}$ and $0 < s_1 \leq s_2$,

$$s_2^{-1}\{\Xi(t) - \xi_0\} \subseteq s_1^{-1}\{\Xi(t) - \xi_0\}.$$

Theorem 3.11. Under Assumptions 2^{*}, 3, 4^{*}, 5^{*}, 6, 7, 8, and 9,

$$\Lambda_n^{(ns)} = \sup_{t \in \mathcal{T}_{ns}} X_n(t) \xrightarrow{\mathcal{L}} \sup_{t \in \mathcal{T}_{ns}} X(t) = \Lambda^{(ns)},$$

where $X(t) = \sup_{\delta \in \Delta(t)} (2\delta^\top Z(t) - \|\delta\|_2^2)$ is defined as in (6) and Z is the limit process from Assumption 5^{*}.

Proof. Let $\tilde{\Lambda}_n^{(ns)} = \sup_{t \in \mathcal{T}_{ns}} \tilde{X}_n(t)$ and $\check{\Lambda}_n^{(ns)} = \sup_{t \in \mathcal{T}_{ns}} \check{X}_n(t)$. For any $\epsilon > 0$, let $\tilde{\Lambda}_n^\epsilon = \sup_{t \in \mathcal{T}_\epsilon} \tilde{X}_n(t)$ and $\check{\Lambda}_n^\epsilon = \sup_{t \in \mathcal{T}_\epsilon} \check{X}_n(t)$. Observe

$$\Lambda_n^{(ns)} = (\Lambda_n^{(ns)} - \tilde{\Lambda}_n^{(ns)}) + (\tilde{\Lambda}_n^{(ns)} - \check{\Lambda}_n^{(ns)}) + \check{\Lambda}_n^{(ns)},$$

with the subtracted quantities finite since $\tilde{\Lambda}_n^{(ns)}, \check{\Lambda}_n^{(ns)} \leq \|Z_n\|_{\mathcal{T}_{ns}}^2 = O_{\mathbf{P}^*}(1)$ by Assumption 5^{*}. Lemma A.3 implies $\Lambda_n^{(ns)} - \tilde{\Lambda}_n^{(ns)} = o_{\mathbf{P}^*}(1)$. Lemma A.5 implies $\check{\Lambda}_n^{(ns)} \xrightarrow{\mathcal{L}} \Lambda^{(ns)}$, so it suffices to show $\tilde{\Lambda}_n^{(ns)} - \check{\Lambda}_n^{(ns)} = o_{\mathbf{P}^*}(1)$.

To that end, let $C_s(t) = s^{-1}\{\Xi(t) - \xi_0\}$, so that $\Delta_n(t) = A(t)^\top C_{n-1/2}(t)$ and $\Delta(t) = A(t)^\top C(t)$. By Assumption 9, if $\delta \in C_{n-1/2}(t)$ then $\delta \in C_{m-1/2}(t)$ for all $m \geq n$. Since $d\{C_{m-1/2}(t), C(t)\} \rightarrow 0$ as $m \rightarrow \infty$ by Assumption 7 and $C(t)$ is closed, this implies $C_{n-1/2}(t) \subseteq C(t)$ and hence $\Delta_n(t) \subseteq \Delta(t)$. Therefore $\tilde{X}_n(t) \leq \check{X}_n(t)$ for all $t \in \mathcal{T}$. Consequently, for every $\epsilon > 0$,

$$0 \leq \tilde{\Lambda}_n^\epsilon \leq \tilde{\Lambda}_n^{(ns)} \leq \check{\Lambda}_n^{(ns)} \quad \text{and} \quad 0 \leq \tilde{\Lambda}_n^\epsilon \leq \check{\Lambda}_n^\epsilon \leq \check{\Lambda}_n^{(ns)}.$$

This gives the bound $0 \leq \check{\Lambda}_n^{(ns)} - \tilde{\Lambda}_n^{(ns)} \leq \check{\Lambda}_n^{(ns)} - \tilde{\Lambda}_n^\epsilon = (\check{\Lambda}_n^{(ns)} - \check{\Lambda}_n^\epsilon) + (\check{\Lambda}_n^\epsilon - \tilde{\Lambda}_n^\epsilon)$. Thus, for

any $\eta > 0$,

$$\mathbf{P}^* \left(|\check{\Lambda}_n^{(ns)} - \tilde{\Lambda}_n^{(ns)}| > \eta \right) \leq \mathbf{P}^* \left(\check{\Lambda}_n^{(ns)} - \check{\Lambda}_n^\epsilon \geq \eta/2 \right) + \mathbf{P}^* \left(\check{\Lambda}_n^\epsilon - \tilde{\Lambda}_n^\epsilon \geq \eta/2 \right).$$

For any fixed $\epsilon > 0$, the second term on the right-hand side tends to zero as $n \rightarrow \infty$ by Lemma A.4. Thus, we are done if we can show that, for any fixed $\alpha > 0$, there is an $\epsilon > 0$ such that the upper limit of the first term on the right-hand side is less than α .

To that end, note Lemma A.5 gives $\check{X}_n \xrightarrow{\mathcal{L}} X$ in $F(\mathcal{T}_{ns})$. Since $\mathcal{T}_\epsilon \subseteq \mathcal{T}_{ns}$, the map $h : F(\mathcal{T}_{ns}) \rightarrow \mathbb{R}$ defined by

$$h(f) = \sup_{t \in \mathcal{T}_{ns}} f(t) - \sup_{t \in \mathcal{T}_\epsilon} f(t)$$

is continuous (in fact, Lipschitz). Therefore, by the continuous mapping theorem,

$$\check{\Lambda}_n^{(ns)} - \check{\Lambda}_n^\epsilon = h(\check{X}_n) \xrightarrow{\mathcal{L}} h(X) = \Lambda^{(ns)} - \Lambda^\epsilon.$$

Moreover, by Assumption 8, for every $t \in \mathcal{T}_{ns}$ we have $d(t, \mathcal{T}_s) > 0$, so $t \in \mathcal{T}_\epsilon$ for all small enough ϵ . Hence $\cup_{\epsilon > 0} \mathcal{T}_\epsilon = \mathcal{T}_{ns}$ and $\Lambda^\epsilon \uparrow \Lambda^{(ns)}$ almost surely as $\epsilon \downarrow 0$. Thus, $\Lambda^{(ns)} - \Lambda^\epsilon \downarrow 0$ almost surely as $\epsilon \downarrow 0$, which implies $\mathbf{P}(\Lambda^{(ns)} - \Lambda^\epsilon \geq \eta/2) \rightarrow 0$ as $\epsilon \downarrow 0$. We can therefore pick an ϵ such that $\mathbf{P}(\Lambda^{(ns)} - \Lambda^\epsilon \geq \eta/2) \leq \alpha$. Then, by the portmanteau theorem [van der Vaart and Wellner, 2023, Theorem 1.3.4],

$$\limsup_{n \rightarrow \infty} \mathbf{P}^* \left(\check{\Lambda}_n^{(ns)} - \check{\Lambda}_n^\epsilon \geq \eta/2 \right) \leq \mathbf{P}(\Lambda^{(ns)} - \Lambda^\epsilon \geq \eta/2) \leq \alpha,$$

which finishes the proof. $\square$

Theorem 3.11 can be used to obtain the asymptotic distribution of Λ_n in different ways. For example, if $\mathcal{T}_s$ is finite and X_n is càdlàg, that is, right-continuous with left limits, then $\Lambda_n = \sup_{t \in \mathcal{T}} X_n(t) = \sup_{t \in \mathcal{T}_{ns}} X_n(t)$ for every n . This follows because $\mathcal{T}_{ns}$ is dense and right-continuity forces the values at points in $\mathcal{T}_s$ to agree with right limits in $\mathcal{T}_{ns}$. Hence, the limit distribution of Λ_n is the same as that of $\Lambda_n^{(ns)}$.

The following corollary gives a general result in this direction. We say $t \mapsto \Xi(t)$ is inner semicontinuous if for every $t \in \mathcal{T}$ and sequence (t_m) tending to t , there exist $\xi_m \in \Xi(t_m)$ such that $\xi_m \rightarrow \xi$ for some $\xi \in \Xi(t)$. When $\mathcal{T}_{ns}$ is dense and $t \mapsto \Xi(t)$ is inner semicontinuous on $\mathcal{T}_{ns}$, it can always be extended to an inner semicontinuous map on $\mathcal{T}$ by defining

$$\Xi(t) = \{\xi \in \mathbb{R}^p : \xi = \lim_{m \rightarrow \infty} \xi_m, \text{ for some } t_m \rightarrow t, \xi_m \in \Xi(t_m)\}, \quad t \in \mathcal{T}_s. \quad (20)$$

Corollary 3.12. *Under the assumptions of Theorem 3.11, suppose additionally that (i) for every n , $\ell_n : \Theta \rightarrow \mathbb{R}$ is lower semicontinuous, (ii) $t \mapsto \Xi(t)$ and $t \mapsto \Delta(t)$ are both inner semicontinuous on $\mathcal{T}_{ns}$, and (iii) Z can be extended to an almost surely continuous process on $\mathcal{T}$. Then $\Lambda_n \xrightarrow{\mathcal{L}} \Lambda = \sup_{t \in \mathcal{T}} X(t)$, where X is defined as in (6) with Ξ and Δ extended to $\mathcal{T}$ using (20).*

4 Independent observations

Suppose $Y_1, \dots, Y_n$ are i.i.d. with density $f(y; \theta)$, with log-likelihood $\ell_n(\theta; \omega) = \sum_{i=1}^n \log f(Y_i(\omega); \theta)$ and score $U_n(t) = n^{-1/2} \sum_{i=1}^n \nabla_\xi \log f(Y_i; \xi_0, t)$, so that $Z_n(t) = A(t)^{-1} U_n(t)$. Define the per-observation cross-information

$$I(t, t') = \mathbb{E}_{\xi_0} \{ \nabla_\xi \log f(Y_1; \xi_0, t) \nabla_\xi \log f(Y_1; \xi_0, t')^\top \}, \quad (21)$$

so that $I(t) = I(t, t)$. Sufficient conditions for Assumption 2 are in Proposition 4.1.

Proposition 4.1. *Suppose (i) $Y_1, Y_2, \dots$ are i.i.d.; (ii) the average observed information $\hat{I}_n(t) = -n^{-1} \nabla_\xi^2 \ell_n(\xi_0, t)$ satisfies $\|\hat{I}_n - I\|_{\mathcal{T}} = o_{p^*}(1)$ with $I(t) = \mathbb{E}_{\xi_0} \{ \hat{I}_n(t) \}$ satisfying Assumption 4; and (iii) all third-order partial derivatives of $\log f(y; \theta)$ with respect to ξ are bounded in a neighborhood of ξ_0 by an integrable $M(y)$, uniformly in θ . Then, with $U_n(t) = n^{-1/2} \sum_{i=1}^n \nabla_\xi \log f(Y_i; \xi_0, t)$, Assumption 2 holds with a remainder term $R_n(\xi, t)$ defined in (3).*

The covariance of Z_n is

$$\rho(t, t') = A(t)^{-1} I(t, t') A(t')^{-\top}, \quad (22)$$

with marginal covariance $\rho(t, t) = \text{Id}_p$. Verification of Assumption 5 reduces to a central limit theorem for Z_n together with asymptotic tightness in $F^p(\mathcal{T})$, the latter typically established by showing the score functions form a Donsker class [van der Vaart and Wellner, 2023, Chapter 2].

Under H_0 , the limit process Z is a centered Gaussian process with covariance ρ . Under a sequence

$$\theta_n = (\xi_0 + n^{-1/2}v, t_0) \quad (23)$$

of contiguous alternatives with $v \neq 0$, Z is non-centered with the same covariance ρ and mean function $\mu(t; t_0, v) := \rho(t, t_0) A(t_0)^\top v$, by Taylor expansion of $f(y; \xi_0 + n^{-1/2}v, t_0)$ and

$A(t)^{-1}I(t, t_0) = \rho(t, t_0)A(t_0)^T$. This extends a result of Davies [1977] from $p = 1$ to general $p \geq 1$.

5 Examples

In this section we illustrate key aspects of our theory with examples. For most of these examples we have a sequence of n i.i.d. random variables $Y_i \in \mathcal{Y}$, with log-likelihood $\ell_n(\theta)$. Moreover, the index set $\mathcal{T} \subset \mathbb{R}^q$ of Z_n and Z will be a subset of q -dimensional Euclidean space for some positive integer q . We further assume in this section that the set $C(t)$, introduced in (11), is a closed, convex cone of the following form: For each $t \in \mathcal{T}$ there is a positive integer $0 \leq r \leq p$ (independent of t) such that each $C(t)$ is determined by the intersection of r half planes, that possibly depend on t . In more detail, for each $t \in \mathcal{T}$ we associate an $r \times p$ matrix $V(t)$ with linearly independent row vectors $v_1(t), \dots, v_r(t)$, such that

$$C(t) = C_{V(t)} =: \{\xi \in \mathbb{R}^p; v_i(t)\xi \geq 0 \text{ for } i = 1, \dots, r\}. \quad (24)$$

From this it follows that $\Delta(t) = A(t)^T C(t) = \{\delta \in \mathbb{R}^p; u_i(t)\delta \geq 0 \text{ for } i = 1, \dots, r\}$, where $u_i(t) = v_i(t)A(t)^{-T}$. Note that $v_i(t)$ can be renormalized by a positive constant without changing $C(t)$ or $\Delta(t)$.

Since $\Delta(t)$ is also a closed convex cone, we deduce from (8) that $X(t) = \|P_{\Delta(t)}Z(t)\|_2^2$ is the squared norm of the projection of $Z(t)$ onto $\Delta(t)$. In order to find the marginal distribution of $X(t)$, assume first that $\xi = \xi_0$. It then follows from Section 4 that Z is a centered Gaussian process with marginal covariance matrix Id_p , and consequently the process X is a $\bar{\chi}^2$ -process with a marginal distribution of $X(t)$ given by (9), where $w_i(t)$ is the probability that the projection $P_{\Delta(t)}Z(t)$ belongs to a face of $\Delta(t)$ with dimension i , see Sen and Silvapulle [2002] and references therein. From this we conclude that $X(t)$ is a mixture of r χ^2 -distributions, corresponding to the r nonzero weights $w_{p-r+1}(t), \dots, w_p(t)$. In particular, when $r = 0$ we have $C(t) = \Delta(t) = \mathbb{R}^p$, so that $w_p(t) = 1$ in (9), and X is a pure χ_p^2 -process. When $r = 1$, $\Delta(t)$ is a half-plane and $X(t)$ is a 0.5 : 0.5-mixture of χ^2 -distributions with $p - 1$ and p degrees of freedom, i.e. $w_{p-1}(t) = w_p(t) = 0.5$. When $r = 2$, $X(t)$ is a mixture of three types of χ^2 -distributions, with weights $w_{p-2}(t) = 0.5 - w_p(t)$, $w_{p-1}(t) = 0.5$ and

$$w_p(t) = \cos^{-1}(-u_1(t)u_2(t)^T / \|u_1(t)\|_2 \|u_2(t)\|_2) / 2\pi, \quad (25)$$

where $u_1(t)$ and $u_2(t)$ are the two row vectors of $U(t)$.

On the other hand, if $\theta_n = (\xi_0 + n^{-1/2}v, t_0)$, then $Z(t) \sim N(\mu(t), \text{Id}_p)$, with $\mu(t) = \mu(t; t_0, v)$. From this it follows that X is a noncentral $\bar{\chi}^2$ -process with marginal distribution $X(t) = \|P_{\Delta(t)}[\mu(t) + \varepsilon(t)]\|_2^2$ and $\varepsilon(t) \sim N(0, \text{Id}_p)$. This distribution is a mixture of truncated, noncentral χ^2 -distributions, whose weights $w_{p-r+1}(t), \dots, w_p(t)$ not only depend on $\Delta(t)$ but also on $\mu(t)$.

Since ℓ_n is the i.i.d. log-likelihood, the random variable Λ_n in (17) is the log-likelihood ratio test statistic for testing the null hypothesis H_0 in (16) against H_1 . Let $0 \leq s \leq p$ be the dimensionality of the null parameter set Ξ_0 in (16). In particular, a simple null hypothesis $\Xi_0 = \{\xi_0\}$ corresponds to $s = 0$.

In the examples of Sections 5.1–5.3 we will write $\Delta(t) = \Delta$, $A(t) = A$, and $C(t) = C$ whenever these quantities are independent of $t \in \mathcal{T}$. For each example we will also give the values of q , p , r and s .

5.1 Mixture models

In this section we give a number of mixture models for which our results can be applied. Rather than a complete verification of all assumptions in each example, which would take substantial space, we at several points refer to existing results for that particular setting. Asymptotic distribution results for the LRT statistic of other mixture models can be found in Bickel and Chernoff [1993], Chernoff and Lander [1995], Lindsay [1995], Lemdani and Pons [1997], Dacunha-Castelle and Gassiat [1999] and references therein.

Example 5.1 (Mixture distributions with $q = p = r = 1$, $s = 0$). Assume, independently for $i \in \{1, \dots, n\}$,

$$Y_i \sim (1 - \xi)N(0, 1) + \xi N(t, 1) \quad (26)$$

with $\mathcal{Y} = \mathbb{R}$, $\mathcal{T} = [-T, T]$ for some $T > 0$ and $\Xi = [0, 1]$. The mixing weight ξ is the parameter of interest, t (the component mean) is unidentifiable under the simple null hypothesis $H_0 : \xi = 0$, so that $s = 0$ and $\xi_0 = 0$ is on the boundary of Ξ . It follows that $p = 1$, $C = \Delta = [0, \infty)$, the per-observation score is $\nabla_\xi \log f(Y_i; \xi_0, t) = \exp(tY_i - t^2/2) - 1$, the cross-information in (21) simplifies to $I(t, t') = \exp(tt') - 1$ (so that $I(t) = I(t, t) = \exp(t^2) - 1$), $\mathcal{T}_s = \{0\}$,

$$Z_n(t) = \frac{1}{\sqrt{n\{\exp(t^2) - 1\}}} \sum_{i=1}^n \{\exp(tY_i - t^2/2) - 1\}$$

if $t \neq 0$ and by taking the limit of $Z_n(t)$ as $t \rightarrow 0$, $Z_n(0) = \sum_{i=1}^n Y_i / \sqrt{n}$. The covariance

function of Z_n and the limit process Z is

$$\rho(t, t') = \{\exp(tt') - 1\} / [\{\exp(t^2) - 1\}\{\exp((t')^2) - 1\}]^{1/2}.$$

The mean function of Z is $\mu(t) = 0$ under H_0 , and it is $\mu(t) = \rho(t, t_0)\sqrt{I(t_0)}v$ under a sequence (23) of contiguous alternatives. Since $\Delta = [0, \infty)$, $X(t) = \max(Z(t), 0)^2$. That $\Lambda_n \xrightarrow{\mathcal{L}} \Lambda$ as $n \rightarrow \infty$ has been established under H_0 by Chen and Chen [2001, Theorem 2]. $\square$

Example 5.2 (Mixture distributions with $q = 2, p = r = 1, s = 0$). We generalize Example 5.1 so that the mean and variance of the second component in (26) are unknown, i.e.,

$$Y_i \sim (1 - \xi)N(0, 1) + \xi N(t_1, t_2),$$

with $\mathcal{Y} = \mathbb{R}$, $t = (t_1, t_2)$ and $\mathcal{T} = [-T_1, T_1] \times [T_{21}, T_{22}]$ for some $T_1 > 0$ and $0 < T_{21} < T_{22} < 2$. In this case $q = 2, p = 1, C = \Delta = [0, \infty)$, and the per-observation score at the simple null hypothesis $\xi_0 = 0$, corresponding to $s = 0$, is

$$\nabla_{\xi} \log f(Y_i; \xi_0, t) = \frac{1}{\sqrt{t_2}} \exp \left\{ \frac{1}{2} (1 - 1/t_2) Y_i^2 + \frac{t_1}{t_2} Y_i - \frac{t_1^2}{2t_2} \right\}.$$

Since $I(t)$ vanishes at $t = (0, 1)$, we have $\mathcal{T}_s = \{(0, 1)\}$, and the theory of Section 3.3 is needed. The restriction $t_2 < 2$ ensures $I(t)$ is finite. As in Example 5.1, $C = \Delta = [0, \infty)$ gives

$$\Lambda = \sup_{t \in \mathcal{T}} \max(0, Z(t))^2,$$

where the covariance function $\rho(t, t')$ of Z is deduced from (21) and (22) on $\mathcal{T}_{ns} \times \mathcal{T}_{ns}$ and then extended to all of $\mathcal{T} \times \mathcal{T}$. The mean function of Z is $\mu(t) = 0$ under H_0 , and it equals $\rho(t, t_0)\sqrt{I(t_0)}v$ under a sequence (23) of contiguous alternatives. $\square$

Example 5.3 (Mixture distributions with $q = r = 1, p = 2, s = 0$ or 1). A second generalization of Example 5.1 is

$$Y_i \sim (1 - \xi_2)N(\xi_1, 1) + \xi_2 N(t, 1),$$

with $\mathcal{Y} = \mathbb{R}$, $\mathcal{T} = [-T, T]$, $\xi = (\xi_1, \xi_2)$ and $\Xi = [-T, T] \times [0, 0.5]$. The restriction $\xi_2 \leq 0.5$ is imposed for identifiability, since $\theta = (t, \xi_1, \xi_2)$ and $\theta' = (\xi_1, t, 1 - \xi_2)$ give the same model. Let us first consider a null parameter that corresponds to a simple null hypothesis $\xi_0 = (0, 0)$

and $s = 0$. The per-observation score at ξ_0 is

$$\nabla_{\xi} \log f(Y_i; \xi_0, t) = [Y_i, \exp(tY_i - t^2/2) - 1]^T,$$

the cross information matrix is

$$I(t, t') = \begin{pmatrix} 1 & t' \\ t & e^{tt'} - 1 \end{pmatrix} \quad (27)$$

and $\mathcal{T}_s = \{0\}$. We use a Cholesky decomposition of $I(t) = I(t, t)$, that is, choose a lower triangular $A(t)$ satisfying $A(t)A(t)^T = I(t)$, defined through its inverse

$$A(t)^{-1} = \begin{pmatrix} 1 & 0 \\ -t/\sigma_t & 1/\sigma_t \end{pmatrix}, \quad (28)$$

with $\sigma_t^2 = e^{t^2} - 1 - t^2$. From the definition $Z_n(t) = A(t)^{-1}U_n(t)$ it follows that $Z_n(t) = [Z_{n1}(t), Z_{n2}(t)]^T$, where $Z_{n1}(t) = \sum_{i=1}^n Y_i/\sqrt{n}$, $Z_{n2}(t) = \sum_{i=1}^n (\exp(tY_i - t^2/2) - 1 - tY_i)/(\sqrt{n}\sigma_t)$ for $t \neq 0$, and by taking the limit of $Z_{n2}(t)$ as $t \rightarrow 0$, $Z_{n2}(0) = \sum_{i=1}^n (Y_i^2 - 1)/\sqrt{2n}$. The covariance function of Z_n and the limit process Z is found by inserting (27) and (28) into (22), giving

$$\rho(t, t') = \begin{pmatrix} 1 & 0 \\ 0 & \sigma_{tt'}/(\sigma_t\sigma_{t'}) \end{pmatrix},$$

where $\sigma_{tt'} = e^{tt'} - 1 - tt'$. Hence, $Z(t) = [Z_1, Z_2(t)]^T$, where $Z_1 \sim N(\mu_1, 1)$ and Z_2 is a real-valued Gaussian process, independent of Z_1 , with mean function $\mu_2(t)$ and covariance function $\sigma_{tt'}/(\sigma_t\sigma_{t'})$. Under H_0 we have $\mu_1 = \mu_2(t) = 0$, whereas

$$\mu(t) = \begin{pmatrix} \mu_1 \\ \mu_2(t) \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & \sigma_{tt_0}/(\sigma_t\sigma_{t_0}) \end{pmatrix} \begin{pmatrix} 1 & t_0 \\ 0 & \sigma_{t_0} \end{pmatrix} \begin{pmatrix} v_1 \\ v_2 \end{pmatrix} = \begin{pmatrix} v_1 + t_0v_2 \\ (\sigma_{tt_0}/\sigma_t)v_2 \end{pmatrix}$$

under a sequence (23) of contiguous alternatives.

Since $\xi_2 \geq 0$, the constraint on ξ implies $C = \{(c_1, c_2)^T : c_2 \geq 0\}$, the upper half plane. Applying $A(t)^T$ using (28), the corresponding set of δ is

$$\Delta = A(t)^TC = \{(\delta_1, \delta_2)^T : \delta_2 \geq 0\}$$

for all $t \in \mathcal{T}$. It follows that $X(t) = \|P_\Delta Z(t)\|_2^2 = Z_1^2 + \max(Z_2(t), 0)^2$, and hence

$$\Lambda = Z_1^2 + \sup_{-T \leq t \leq T} \max(Z_2(t), 0)^2.$$

That $\Lambda_n \xrightarrow{\mathcal{L}} \Lambda$ has been proved under H_0 by Chen and Chen [2001, Theorem 3].

To illustrate Theorem 3.9, we consider another (composite) null $H_0 : \xi_2 = 0, \xi_1$ free, which tests whether the two-component mixture reduces to a single Gaussian with unknown mean. This corresponds to a composite null hypothesis, more specifically a null parameter space $\Xi_0 = [-T, T] \times \{0\}$ and $s = 1$. Under H_0 , the model is $Y_i \sim N(\xi_1, 1)$ regardless of t , so t is unidentifiable under H_0 . Optimizing over ξ_1 gives $X_n^0(t) = 2 \sup_{\xi_1 \in [-T, T]} \{\ell_n(\xi_1, 0, t) - \ell_n(0, 0, t)\} = n\bar{Y}_n^2$, which does not depend on t and converges in law to Z_1^2 . Thus $X^0(t) = Z_1^2$ is a constant process, and $\|X_n^0\|_{\mathcal{T}} \xrightarrow{\mathcal{L}} Z_1^2$. By Theorem 3.9 and the limit from the simple null above,

$$\Lambda_n \xrightarrow{\mathcal{L}} \Lambda = [Z_1^2 + \sup_{-T \leq t \leq T} \max(Z_2(t), 0)^2] - Z_1^2 = \sup_{-T \leq t \leq T} \max(Z_2(t), 0)^2, \quad (29)$$

the same limit as in Example 5.1. The Z_1^2 contribution from the identifiable mean parameter ξ_1 cancels exactly. Note also that the limiting distribution of Λ in (29) holds both under the null hypothesis, as well as under a sequence (23) of contiguous alternatives. For asymptotics under H_0 , this is the known-variance special case of Chen and Chen [2003], who derive the same limit for the composite null in a normal mixture with unknown variance σ^2 ; the additional χ_1^2 contribution from estimating σ^2 cancels by the same mechanism. $\square$

5.2 Linkage analysis and MOD scores

It has been noticed that some test statistics arising in genetic linkage analysis can be formulated as the supremum of a one-dimensional Gaussian or χ_1^2 -process along an interval, see, for instance, Lander and Botstein [1989] and Feingold et al. [1993]. Dupuis et al. [1995] define another test statistic that asymptotically equals the supremum of a χ_2^2 -process along an interval. In this section, we show that LR-tests in linkage analysis, usually referred to as MOD scores (maximized over disease models), fit into our framework.

In order to present these asymptotic LR-test results, let us start by briefly summarizing the genetic linkage model; see Sham [1998] for more details. The objective is to test whether the position (or locus) τ of a disease susceptibility gene is along a given chromosome $[0, T]$ of

genetic map length T :

$$\begin{aligned} H_0 : \tau &\notin [0, T], \\ H_1 : \tau &\in [0, T]. \end{aligned} \tag{30}$$

At our disposal we have n families, with information on how the disease (phenotypes) as well as DNA is segregated in each family.

5.2.1 One family type

In this section we assume that all n families are of the same type. By this we mean that not only the number N of individuals is the same for each family, but also the pedigree structure $\mathcal{P}$ and the phenotype vector $\Phi = (\Phi_k, k \in \mathcal{P})$. Transmission of genetic material at position $s \in [0, T]$ along the chromosome is described by a binary inheritance vector $v(s) = (v_1(s), \dots, v_m(s)) \in \{0, 1\}^m$, where m is the total number of meioses and $v_l(s) \in \{0, 1\}$ indicates whether grandpaternal or grandmaternal DNA was transmitted during the l th meiosis, with switches between these two states corresponding to points of crossovers. There are two meioses (one from each parent) for each nonfounder, so that $m/2$ and $N - m/2$ is the number of nonfounders and founders of the pedigree, respectively. The complete marker data $Y = \{v(s); 0 \leq s \leq T\} \in \mathcal{Y}$ is observed for each family. Here $\mathcal{Y}$ is the space of functions $y : [0, T] \rightarrow \{0, 1\}^m$ with at most a finite number of jumps, corresponding to the total number of crossovers for all meioses of the pedigree.

We will restrict ourselves to binary phenotypes, so that an individual is either unaffected or affected by a disease. Consider a biallelic disease gene and assume that the allele that increases the risk of being affected has frequency $p \in [0, 1]$ in the population. Define penetrance parameters $\gamma = (\gamma_0, \gamma_1, \gamma_2) \in [0, 1]^3 = \Gamma$, where γ_j is the probability for an individual to be affected given j copies of the disease allele (that is, j corresponds to the number of parents that pass on the disease allele to this individual). With $\theta = (\tau, p, \gamma)$, the density of Y given Φ is

$$f(y; \theta) = P_{p, \gamma}(v(\tau) | \Phi) P(y | v(\tau)). \tag{31}$$

The exact form of the first term $P_{p, \gamma}(v(\tau) | \Phi)$ on the right-hand side of (31) is in Section A.2 of the Supplementary material. Assuming Mendelian inheritance, the null hypothesis in (30) corresponds to a uniform distribution

$$P_{p, \gamma}(v(\tau) | \Phi) = 2^{-m} \tag{32}$$

of $v(\tau)$ over $\{0, 1\}^m$. If additionally Haldane's map function for crossovers is assumed, all

$\{v_l(s); 0 \leq s \leq T\}$ are independent Markov processes on $\{0, 1\}$ for $l = 1, \dots, m$, with intensity 1 for each process to switch between its two states (when genetic map length along the chromosome is measured in Morgans). This implies that $P(y|v(\tau)) = \exp(-mT)$, independently of the total number of crossovers for all m meioses, and independently of where along the chromosomes these crossovers are located. Since $P(y|v(\tau))$ does not depend on τ it is a constant that can be dropped from the per-pedigree likelihood $f(y; \theta)$.

To bring the model into the framework of the present article, we reparametrize the penetrance parameters. For fixed p , we introduce an inner product and an associated orthonormal basis $e_0 = (1, 1, 1), e_1, e_2$ for $\mathbb{R}^3$ depending on p (see Section A.2 of the Supplementary material for the definition of the scalar product of $\mathbb{R}^3$, e_1 and e_2). Write

$$\gamma = K e_0 + \sqrt{\xi_1} e_1 + \sqrt{\xi_2} e_2, \quad (33)$$

where $K = E(\Phi_k)$ is the disease prevalence, and $\xi_1 = \varepsilon_1^2$, $\xi_2 = \varepsilon_2^2$ are the additive and dominance components of genetic variance. We set

$$t = (\tau, p, K), \quad \xi = (\xi_1, \xi_2), \quad (34)$$

with $\mathcal{T} = [0, T] \times [0, 1]^2$ and $\Xi(t) = \{\xi \in [0, \infty)^2 : \gamma(\xi; K, p) \in \Gamma\}$, corresponding to $q = 3$ and $p = 2$ respectively. Since the null hypothesis in (30) is mathematically equivalent to (32), under certain conditions it can also be formulated in terms of the two genetic variance components $\xi = (\xi_1, \xi_2)$. In order to demonstrate this, we introduce for each p the set $\Gamma_0(p) \subset \Gamma$ of penetrance vectors γ for which (32) holds. We will assume that the pedigree type satisfies

$$\Gamma_0(p) = \{\gamma : \gamma_0 = \gamma_1 = \gamma_2\} \text{ for all } p. \quad (35)$$

In general $\Gamma_0(p)$ is at least as large as the right-hand side of (35). For instance, if the phenotype of at most one individual of the pedigree is known, then $\Gamma_0(p) = \Gamma$ for all p and (35) fails. Condition (35) will however hold for most pedigree types of interest, and in view of (33) it implies that the null hypothesis condition (32) is equivalent to a simple null hypothesis $\xi_0 = (0, 0)$ for the genetic variance vector ξ , corresponding to a simple null hypothesis and $s = 0$. This is to say that testing whether the disease gene of interest is located along the chromosome $[0, T]$ or not is mathematically equivalent to testing whether ξ equals ξ_0 or not.

The per-observation score $\nabla_\xi \log f(y; \xi_0, t) = S(v(\tau); K)$ was derived in Hössjer [2005a], based on methods developed in Hössjer [2003]. Since $p = 2$, the per-observation score vector

$S = (S_1, S_2)^\top$ has two components given by

$$S_1(v) = \sum_{1 \leq k < l \leq N} \omega_{kl} \text{IBD}_{kl}(v)/2 - C_1; \quad S_2(v) = \sum_{1 \leq k < l \leq N} \omega_{kl} \mathbf{1}_{\{\text{IBD}_{kl}(v)=2\}} - C_2, \quad (36)$$

where $\text{IBD}_{kl}(v) \in \{0, 1, 2\}$ is the number of alleles from the founders of the family that are shared identical-by-descent by k and l from the same founder alleles, C_i is a centering constant assuring that $\mathbb{E}_{\xi_0}\{S_i(v)\} = 0$, whereas ω_{kl} is a weight assigned to each pair (k, l) of individuals, based their phenotypes (see Section A.2 of the Supplementary material). For binary phenotypes we assign $\Phi_k = 0, 1$ or $?$, depending on whether k is affected, unaffected or has unknown phenotype. The weights of pair (k, l) then simplify to

$$\omega_{kl} = \begin{cases} K^{-2}, & \Phi_k = \Phi_l = 1, \\ -K^{-1}(1 - K)^{-1}, & \Phi_k \neq \Phi_l, \Phi_k, \Phi_l \in \{0, 1\}, \\ (1 - K)^{-2}, & \Phi_k = \Phi_l = 0, \\ 0, & \text{otherwise.} \end{cases} \quad (37)$$

Since we restrict ourselves to binary phenotypes, we assume that (37) holds. It follows from (21) that the cross information matrix is

$$\begin{aligned} I(t, t') &= \mathbb{E}_{\xi_0} [S(v(\tau); K) S(v(\tau'); K')^\top] \\ &= \sum_{v, v'} S(v; K) S(v'; K')^\top \mathbb{P}_{\xi_0}(v(\tau) = v, v(\tau') = v') \end{aligned} \quad (38)$$

between t and $t' = (\tau', p', K')$. The last sum ranges over all 4^m pairs of inheritance vectors v, v' , under the null hypothesis that all components of $Y = \{v_i(s); 0 \leq s \leq T\}_{i=1}^m$ are independent Markov processes on $\{0, 1\}$ with intensity 1 of jumping between their two states. It follows from (36) and (37) that the per-observation score only depends on t through τ and K . Hence we deduce from (38) that the cross information matrix (38) is only a function of (τ, K) and (τ', K') . Provided the pedigree type satisfies (35), and if $\epsilon \leq K \leq 1 - \epsilon$ for some $\epsilon > 0$, it follows that the per individual Fisher information matrix $I(t) = I(t, t)$ is nonsingular for all $t \in \mathcal{T}$, i.e. $\mathcal{T}_{ns} = \mathcal{T}$.

Although $\log f(y; \theta)$ is not twice differentiable in ξ at ξ_0 owing to the reparametrization (33), the required quadratic expansion of the log-likelihood still holds as the number n of families grows:

Theorem 5.4 (Quadratic log-likelihood expansion for linkage.). *Assume marker data Y_i of*

families $i = 1, \dots, n$ are independent. The log-likelihood, with $f(Y_i; \theta)$ as in (31), then admits a quadratic expansion of the form (3), with $U_n(t)$ as in Section 4, $I(t) = I(t, t)$ as in (38), $A(t)$ satisfying $A(t)A(t)^\top = I(t)$, and a remainder R_n satisfying Assumption 2.

In order to find the asymptotic distribution Λ of the likelihood ratio test statistic under the simple null hypothesis H_0 we use Theorem 3.1. Since $\xi_1 \geq 0$ and $\xi_2 \geq 0$, there are $r = 2$ boundary restrictions with $C(t) = C = [0, \infty)^2$ a convex, closed cone, and therefore $\Delta(t) = A(t)^\top C$ is a closed, convex cone as well. Consequently, $X(t) = \|P_{\Delta(t)}Z(t)\|_2^2$, with Z a centered Gaussian process with covariance function $\rho(t, t')$ given by (22). Since $I(t, t')$ only depends on t and t' through (τ, K) and (τ', K') , the same is true for $\rho(t, t')$, and therefore the limit process $X(t)$ will not depend on p . Hence we regard $X(\tau, K)$ as a function of (τ, K) only, and the asymptotic LRT statistic reduces to $\Lambda = \sup_{\tau \in [0, T], K \in [\epsilon, 1-\epsilon]} X(\tau, K)$. There are $r = 2$ constraints that define $\Delta(t)$. From this and (9) it follows that

$$X(\tau, K) \sim (0.5 - w_2(\tau, K))\chi_0^2 + 0.5\chi_1^2 + w_2(\tau, K)\chi_2^2, \quad (39)$$

has a $\bar{\chi}^2$ -distribution under H_0 for each (τ, K) , with

$$w_2(\tau, K) = \cos^{-1}(I_{12}(\tau, K)/\sqrt{I_{11}(\tau, K)I_{22}(\tau, K)})/(2\pi). \quad (40)$$

The weight (40) of the χ_2^2 component in (39) can be deduced from (25). Indeed, since $C = [0, \infty)^2$, we may take $V(t) = \text{Id}_2$ in (24), giving $u_i(t) = [A(t)^{-\top}]_i$ (the i -th row of $A(t)^{-\top}$). Then $u_i(t)u_j(t)^\top = [I(t)^{-1}]_{ij} = I_{ij}^{-1}(\tau, K)$, and consequently

$$\frac{u_1(t)u_2^\top(t)}{\|u_1(t)\|_2\|u_2(t)\|_2} = \frac{I_{12}^{-1}(\tau, K)}{\sqrt{I_{11}^{-1}(\tau, K)I_{22}^{-1}(\tau, K)}} = -\frac{I_{12}(\tau, K)}{\sqrt{I_{11}(\tau, K)I_{22}(\tau, K)}}.$$

Example 5.5 (MOD scores with only affecteds.). When all pedigree members with known phenotype are affected, (36)-(37) simplify to

$$S(v) = K^{-2}(S_{\text{pairs}}(v) - \mathbb{E}_0\{S_{\text{pairs}}\}, S_{\text{g-prs}}(v) - \mathbb{E}_0\{S_{\text{g-prs}}\})^\top, \quad (41)$$

where $S_{\text{pairs}} = \sum_{k < l} \text{IBD}_{kl}/2$ [Whittemore and Halpern, 1994], and $S_{\text{g-prs}} = \sum_{k < l} \mathbf{1}_{\{\text{IBD}_{kl}=2\}}$ [McPeck, 1999], with sums over all affected pairs (k, l) . In this case the limit process $Z(t) = Z(\tau)$ under H_0 is a function of τ only. It is a two-dimensional stationary Gaussian process on $[0, T]$ with

$$\rho(\tau, \tau') = \rho(\tau' - \tau). \quad (42)$$

It is shown in Section A.2 of the Supplementary material that the covariance function has the form

$$\rho(s) = \sum_{l=1}^m \kappa_l \exp(-2l|s|), \quad (43)$$

corresponding to a stationary diffusion; where κ_l are 2×2 matrices given in Table 1 for the pedigree types of Figure 1. For affected sibling pedigrees (types 1–4), the two components Z_1 and Z_2 of Z are independent Ornstein-Uhlenbeck processes along $[-T, T]$. The asymptotic LR-statistic then simplifies to

$$\Lambda = \sup_{0 \leq \tau \leq T} X(\tau), \quad (44)$$

with $X(\tau) = \|P_\Delta Z(\tau)\|_2^2$ a stationary $\bar{\chi}^2$ -process that is a mixture (39) between χ^2 -distributions with 0, 1 and 2 degrees of freedom. Since K^{-2} enters as a multiplicative constant in (41), it follows that $I(t) = I(t, t) = K^{-4}I_0$, for some 2×2 matrix I_0 independent of t . Hence we deduce from (40) that the weight w_2 of the χ_2^2 -distribution is a constant, independent of t . Figure 1 shows several pedigree types with only affecteds; the corresponding $I(t)$, $\Delta = C_U$, and w_2 are in Table 2 of Section A.2 of the Supplementary material. $\square$

Example 5.6 (Unilineal pairs). An affected unilineal pair (k, l) is a pedigree type in which both k and l are affected, all other members have unknown phenotype, and (k, l) can share at most one allele IBD (for instance, affected first cousins or uncle-nephew pairs, but not affected sib pairs). For such pedigrees, the above parametrization $\theta = (\xi, t)$ with two genetic variance components in ξ does not work, since (35) is not satisfied. For this reason $I(t)$ is singular for *all* values of t , and therefore the framework of Section 3.3 does not apply. Instead another parametrization $\theta = (t, \xi)$ is needed with only one genetic variance component in ξ . This means that $\mathcal{T} \subset \mathbb{R}^4$, $\Xi(t) \subset \mathbb{R}$, $p = 1$, and $\Delta(t) = [0, \infty)$. It can be shown that $X(t)$ is only a function of t through $\tau \in [-T, T]$. Therefore, the asymptotic LRT statistic satisfies (44), the supremum of a $\bar{\chi}^2$ -process with marginal distribution $X(\tau) \sim 0.5 \chi_0^2 + 0.5 \chi_1^2$. Formally, this corresponds to a weight $w_2 = 0$ for the χ_2^2 -component of $X(\tau)$. $\square$

Example 5.7 (Affected sib pairs and MLS scores). For affected sib pairs (pedigree type 1 in Figure 1), there is an alternative parametrization $\theta = (\tau, \xi)$ with $\xi = (z_0, z_1)$, where z_i is the probability that the sib pair shares i alleles IBD at τ [Suarez et al., 1978]. The resulting test, the MLS score, is asymptotically equivalent to the corresponding MOD score; a detailed verification is in Section A.2 of the Supplementary material. $\square$

5.2.2 Several family types

Let us generalize the model of Section 5.2.1 and assume that the $n = n_1 + \dots + n_J$ families are of $J > 1$ types $(\mathcal{P}_1, \Phi_1), \dots, (\mathcal{P}_J, \Phi_J)$ with n_j families of type j , with a density $f_j(y; \theta)$ of marker data for pedigree type $(\mathcal{P}_j, \Phi_j)$. The combined log-likelihood $\ell_n(\theta; \omega) = \sum_{j=1}^J \sum_{i=1}^{n_j} \log f_j(Y_{ji}(\omega); \theta)$ for the whole data set with all J family types generalizes the single-type case, with $Y_{ji} \in \mathcal{Y}_j$ marker data for family i of type j . This log-likelihood admits the same type of asymptotic expansion as in Theorem 5.4, with asymptotic per-observation cross information matrix $I(t, t') = \sum_{j=1}^J \beta_j I_j(t, t')$, where $\beta_j = \lim_{n \rightarrow \infty} n_j/n$ are the asymptotic family type proportions and $I_j(t, t')$ is defined as in (38) for pedigree type $(\mathcal{P}_j, \Phi_j)$.

Example 5.8 (MOD score with affected sib pairs and first cousins.). Put $J = 2$ with asymptotic proportions $\beta_1 = \beta$ of affected sib pairs and $\beta_2 = 1 - \beta$ of affected first cousins. Then

$$I(t) = K^{-4} \begin{pmatrix} 0.125\beta + 0.0469(1 - \beta) & 0.125\beta \\ 0.125\beta & 0.1875\beta \end{pmatrix}.$$

The χ_2^2 -weight w_2 from (40) is plotted as a function of β in Figure 2 of Section A.2 of the Supplementary material. It is notable that $w_2 \rightarrow 0.25$ as $\beta \rightarrow 0^+$, whereas $\beta = 0$ corresponds to the degenerate case of only having unilineal first cousin pairs (Example 5.6) with $w_2 = 0$. This discontinuity suggests that very large sample sizes are needed when β is positive but small for Λ_n to be distributed approximately as (44). $\square$

5.3 Polar-coordinates example

In this section we illustrate that the weak limit of the LRT statistic is the supremum of a Gaussian process also when all parameters are identifiable under the null hypothesis for the (Cartesian) parameter η . This is achieved by switching to polar coordinates. For simplicity we confine ourselves to a two-dimensional parameter, although the results hold more generally. That is, we assume that the common density of the i.i.d. random variables $Y_1, \dots, Y_n$ is $f(y; \eta)$ for some unknown parameter $\eta = (\eta_1, \eta_2)^T \in \mathbb{R}^2$. We wish to test a simple null hypothesis $H_0 : \eta = \eta_0$ against $H_1 : \eta \neq \eta_0$. We will first assume that $\eta_0 = (\eta_{01}, \eta_{02})^T$ is an inner point of the parameter space. Consider a sequence of parameter vectors

$$\eta_n = \eta_0 + n^{-1/2}v[\cos(t_0), \sin(t_0)]^T =: \eta_0 + n^{-1/2}vJ(t_0), \quad (45)$$

for some fixed $v \geq 0$ and $t_0 \in [0, 2\pi]$. Note that $\eta_n = \eta_0$ if $v = 0$, whereas η_n corresponds to a contiguous sequence of alternatives when $v > 0$. We will use bars to denote quantities for this model, which corresponds to $q = s = r = 0$ and $p = 2$. Thus $\bar{\ell}_n(\eta)$ is the i.i.d. log-likelihood and $\bar{U}_n = n^{-1/2} \nabla_{\eta} \bar{\ell}_n(\eta)$ is the score for η at η_0 . Let $\bar{I} = (\bar{I}_{ij})_{i,j=1}^2$ be the 2×2 Fisher information matrix for η at $\eta = \eta_0$, and $\bar{Z}_n = \bar{A}^{-1} \bar{U}_n$ the standardized score, with $\bar{A} = \bar{I}^{1/2}$ the symmetric square root of $\bar{I}$. Under the conditions of Corollary 3.3, $\bar{Z}_n \xrightarrow{\mathcal{L}} \bar{Z} = (\bar{Z}_1, \bar{Z}_2)^{\top} \sim N(\bar{\mu}, \text{Id}_2)$, where $\bar{\mu} = v \bar{A} J(t_0)$, and consequently

$$\Lambda_n \xrightarrow{\mathcal{L}} \Lambda = \|\bar{Z}\|_2^2 \sim \chi_{2, \|\bar{\mu}\|_2^2}^2 \quad (46)$$

as $n \rightarrow \infty$, so that Λ has a noncentral chisquare distribution when $v > 0$, with noncentrality parameter $\|\bar{\mu}\|_2^2 = v^2 J(t_0)^{\top} \bar{I} J(t_0)$.

In polar coordinates we reformulate the model as

$$\eta = \eta_0 + \xi [\cos(t), \sin(t)]^{\top} = \eta_0 + \xi J(t) \quad (47)$$

for $t \in \mathcal{T} = [0, 2\pi]$, $\xi \in \Xi = [0, \infty)$ and $\xi_0 = 0$. After this reparametrization the null hypothesis is still simple ($s = 0$), but now there is a nuisance parameter t of dimension $q = 1$, the parameter ξ of interest is of dimension $p = 1$, and on the boundary of the parameter space ($r = 1$). In polar coordinates, the sequence of parameter values in (45) corresponds to $t = t_0$ and $\xi_n = \xi_0 + n^{-1/2}v$.

In order to verify that the distribution of Λ in (46) is still the same for polar coordinates, we remove the bars for quantities that are evaluated under the polar coordinates parametrization. The Fisher information for ξ at ξ_0 is given by $I(t) = J(t)^{\top} \bar{I} J(t)$, so that $\mathcal{T}_s = \emptyset$ if $\bar{I}$ is nonsingular. Since $p = 1$ and $I(t)$ is scalar, $A(t) = I(t)^{1/2}$ is scalar as well. The standardized score $Z_n(t) = A(t)^{-1} U_n(t)$ for polar coordinates can be written as

$$Z_n(t) = I(t)^{-1/2} U_n(t) = I(t)^{-1/2} J(t)^{\top} \bar{U}_n = I(t)^{-1/2} J(t)^{\top} \bar{A} \bar{Z}_n =: K(t)^{\top} \bar{Z}_n.$$

It is easy to see that $K(t)^{\top} K(t) = \|K(t)\|_2^2 = 1$, and for this reason we can write $K(t) = I(t)^{-1/2} \bar{A} J(t) = J(g(t))$, where $g : [0, 2\pi] \rightarrow [0, 2\pi]$ is a transformed angle. Hence the weak limit of Z_n is a process $Z \in F([0, 2\pi])$ satisfying

$$Z(t) = K(t)^{\top} \bar{Z} = J(g(t))^{\top} \bar{Z},$$

a time-transformed sinusoidal Gaussian process with covariance function $\rho(t, t') = \cos[g(t) - g(t')]$.

$g(t')]$ and mean function

$$\mu(t) = K(t)^\top \bar{\mu} = vI(t_0)^{1/2} K(t)^\top K(t_0) = vI(t_0)^{1/2} \cos[g(t) - g(t_0)],$$

in agreement with $\mu(t; t_0, v) = \rho(t, t_0)A(t_0)^\top v$. Note in particular that Z is a non-centered (centered) Gaussian process when $v > 0$ ($v = 0$) respectively. Since $C = \Delta = [0, \infty)$, we get

$$X(t) = \max(Z(t), 0)^2 = \max[\bar{Z}_1 \cos(g(t)) + \bar{Z}_2 \sin(g(t)), 0]^2.$$

Thus, using that g is a bijection on $[0, 2\pi]$,

$$\Lambda = \sup_{0 \leq t \leq 2\pi} X(t) = \bar{Z}_1^2 + \bar{Z}_2^2 \sim \chi_{2, \|\bar{\mu}\|_2^2}^2, \quad (48)$$

in accordance with the Cartesian-coordinates result. The last equality follows by picking $g(t)$ so that the Cauchy–Schwarz inequality holds with equality ($|K(t)^\top \bar{Z}| \leq \|K(t)\|_2 \|\bar{Z}\|_2$, with equality if $K(t) = \bar{Z}/\|\bar{Z}\|_2$, which is achievable since g is a bijection).

Suppose instead that the original, untransformed parameter space is $\{\eta; \eta_i \geq \eta_{i0} \text{ for } i = 1, 2\}$. The null parameter η_0 is then at the boundary of this parameter space, giving rise to a testing problem with $p = r = 2$ and $q = s = 0$. The neighbourhood of η_0 is a closed convex cone $\bar{C} = [0, \infty)^2$ and the standardized convex cone is $\bar{\Delta} = \bar{A}\bar{C}$. The transformed, polar parameter space, however, is almost the same as before, except that the parameter set of the nuisance parameter changes to $\mathcal{T} = [0, \pi/2]$. It follows, analogously to (46) and (48), that

$$\Lambda = \|P_{\bar{\Delta}} \bar{Z}\|_2^2 = \sup_{0 \leq t \leq \pi/2} X(t). \quad (49)$$

The distribution of Λ when $v > 0$ is a mixture of truncated, noncentral χ^2 -distributions. Under the null hypothesis $v = 0$, the distribution of Λ simplifies to a mixture

$$\Lambda \stackrel{v=0}{\sim} (1 - w_2)\chi_0^2 + 0.5\chi_1^2 + w_2\chi_2^2 \quad (50)$$

of three chisquare distributions. The weight w_2 of the χ_2^2 -distribution can be obtained from (25). Indeed, since $\bar{C} = [0, \infty)^2$ we can choose $\bar{V} = \text{Id}_2$ and $\bar{U} = \bar{V}\bar{A}^{-1} = \bar{I}^{-1/2}$. Letting $\bar{u}_1$ and $\bar{u}_2$ be the two rows of $\bar{U}$ we find from (25) that

$$w_2 = \frac{\cos^{-1}(-\bar{u}_1 \bar{u}_2^\top / \|\bar{u}_1\|_2 \|\bar{u}_2\|_2)}{2\pi} = \frac{\cos^{-1}(-\bar{I}_{12}^{-1} / \sqrt{\bar{I}_{11}^{-1} \bar{I}_{22}^{-1}})}{2\pi} = \frac{\cos^{-1}(\bar{I}_{12} / \sqrt{\bar{I}_{11} \bar{I}_{22}})}{2\pi}. \quad (51)$$

6 Discussion

The limiting distribution of the LRT statistic under the null hypothesis, or a sequence of contiguous alternatives, is the supremum $\Lambda = \sup_{t \in \mathcal{T}} X(t) = \|X\|_{\mathcal{T}}$ of a (noncentral) $\bar{\chi}^2$ -process over $\mathcal{T}$, the set of possible values of the nuisance parameter t that is unidentifiable under the null hypothesis. This is achieved in settings not covered by existing theory. Classical results for boundary inference without nuisance parameters [Chernoff, 1954, Self and Liang, 1987, Geyer, 1994], testing whether a single parameter is zero or positive in the presence of nuisance parameters [Davies, 1977, 1987, Ritz and Skovgaard, 2005], and testing whether a density is a mixture of two components or not [Chen and Chen, 2001] are obtained as special cases. The limiting distribution has the same form whether or not the information is singular.

For composite null hypotheses $H_0 : \xi \in \Xi_0$, Theorem 3.9 shows that the limiting distribution of the LRT statistic is $\|X - X^0\|_{\mathcal{T}}$, where X and X^0 are the (noncentral) $\bar{\chi}^2$ -processes corresponding to the full model and null hypothesis, respectively. This implies that identifiable components of the null hypothesis cancel exactly. When the tangent cone of Ξ_0 is a linear subspace, Theorem 3.10 further identifies the limiting distribution of the LRT statistic as $\Lambda = \sup_{t \in \mathcal{T}} \|P_{K(t)}Z(t)\|_2^2$, a supremum of (noncentral) $\bar{\chi}^2$ variables $X(t) = \|P_{K(t)}Z(t)\|_2^2$ determined by the active boundary constraints. This is illustrated for the mixture problem of Example 5.3 and recovers the known-variance case of Chen and Chen [2003]. In settings where a parameter is unidentifiable under the null in one parametrization but not another, the limit is the same regardless: Section 5.3 illustrates this in a setting where reparameterizing in terms of polar coordinates introduces an unidentifiable nuisance parameter, yet the limiting distribution of the LRT statistic is unchanged.

Our framework includes the number p of parameters to be tested, the number r of boundary constraints on these parameters, as well as the dimensionalities q and s of the nuisance parameters that are not identifiable and identifiable under the null hypothesis, respectively. Related results involving maxima of (noncentral) $\bar{\chi}^2$ -processes but without a general choice of p , r , q and s include Ritz and Skovgaard [2005] ($p = r = 1$, arbitrary q, s), Andrews and Ploberger [1995] ($r = 0$, general p, q, s), and Andrews [1996] ($q = 0$, general p, r, s). The set $\mathcal{T}$ of possible values of the nuisance parameter vector that is unidentifiable under the null hypothesis could be a functional space ($q = \infty$), although $\mathcal{T}$ is a subset of a finite-dimensional Euclidean space ($q < \infty$) in all our examples. Our examples further have $p \leq 2$; cases with $p > 2$, such as mixtures with more than two components or linkage analysis with two disease genes, fit our framework but involve weights of the (noncentral) $\bar{\chi}^2$ -process that are often harder to characterize explicitly.

Although all our examples are based on likelihoods for independent data (and most often likelihoods for i.i.d. data), our framework is much more general than this. The most important requirement for our asymptotic results to hold is weak convergence of the LRT statistic towards the supremum of a stochastic process, regardless of whether data exhibit serial correlations or not.

As mentioned in Section 1, there are many other applications of the framework of the present article than mixture models and genetic linkage analysis. For several of these applications, it would be interesting to characterize the limiting (noncentral) $\bar{\chi}^2$ -process explicitly. Change-point testing [Andrews and Ploberger, 1995] is one such application, with t the change point and ξ the level change after the change point. Unidentifiability under the null hypothesis of no change point follows immediately, and our general theory does not require i.i.d. data to analyze the asymptotic behavior of the LRT statistic for this model.

A systematic treatment, including verification of the assumptions of Section 3 for triangular-array models and characterization of the (noncentral) $\bar{\chi}^2$ -process limit when ξ_0 lies on the boundary of Ξ , is another promising direction for future work. In a related preprint, Cox [2022] develops a generalized argmax theorem that handles structural break estimation and boundary inference using Painlevé–Kuratowski set convergence. These results may provide useful tools for the program of the present article.

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Supplementary material

A.1 Proofs

Proof of Theorem 3.1. By lemmas 3.5, 3.6, and 3.7, $\|X_n - \tilde{X}_n\|_{\mathcal{T}} = o_{p^*}(1)$, $\|\tilde{X}_n - \check{X}_n\|_{\mathcal{T}} = o_{p^*}(1)$, and $\check{X}_n \xrightarrow{\mathcal{L}} X$, respectively. Thus, by Slutsky's lemma [van der Vaart and Wellner, 2023, Example 1.4.7], $(X_n - \tilde{X}_n, \tilde{X}_n - \check{X}_n, \check{X}_n) \xrightarrow{\mathcal{L}} (0, 0, X)$, and hence by continuous mapping theorem [van der Vaart and Wellner, 2023, Theorem 1.3.6], $X_n = (X_n - \tilde{X}_n) + (\tilde{X}_n - \check{X}_n) + \check{X}_n \xrightarrow{\mathcal{L}} 0 + 0 + X = X$. Note the requirement that X be separable holds because X is tight and $F(\mathcal{T})$ a metric space [van der Vaart and Wellner, 2023, p.15]. $\square$

For any closed $A, A' \subseteq \mathbb{R}^p$ and $\bar{B}_M = \{z \in \mathbb{R}^p : \|z\|_2 \leq M\}$, the closed ball centered at the origin with radius $M > 0$, define

$$d_M(A, A') = \max \left(\sup_{z \in A \cap \bar{B}_M} \|z - P_{A'}(z)\|_2, \sup_{z' \in A' \cap \bar{B}_M} \|z' - P_A(z')\|_2 \right)$$

and

$$d(A, A') = \sum_{M=1}^{\infty} 2^{-M} \min\{d_M(A, A'), 1\}. \quad (\text{A.1})$$

Then d defines a metric between closed sets. See for example Geyer [1994] for the connection between Assumption 7 and Chernoff regularity [Chernoff, 1954].

Lemma A.1. *Under Assumptions 4, 6, and 7,*

$$\lim_{n \rightarrow \infty} \sup_{t \in \mathcal{T}} d(\Delta_n(t), \Delta(t)) = 0.$$

Moreover, for any $M > 0$,

$$\lim_{n \rightarrow \infty} \sup_{t \in \mathcal{T}, \|z\|_2 \leq M} \left| \|z - P_{\Delta_n(t)} z\|_2^2 - \|z - P_{\Delta(t)} z\|_2^2 \right| = 0$$

and

$$\lim_{n \rightarrow \infty} \sup_{t \in \mathcal{T}, \|z\|_2 \leq M} \left| \|P_{\Delta_n(t)} z\|_2^2 - \|P_{\Delta(t)} z\|_2^2 \right| = 0.$$

Proof. For the first claim it suffices to prove that for any $M > 0$,

$$\lim_{n \rightarrow \infty} \sup_{t \in \mathcal{T}} d_M(\Delta_n(t), \Delta(t)) = 0.$$

Let $C_s(t) = \{\Xi(t) - \xi_0\}/s$ and pick $z \in \Delta_n(t) \cap \bar{B}_M$ and $z' \in \Delta(t) \cap \bar{B}_M$. Then, by definition of these sets, $z = A(t)^\top y$ for a $y \in C_{n^{-1/2}}(t)$ and $z' = A(t)^\top y'$ for a $y' \in C(t)$. Thus, $\|y\|_2, \|y'\|_2 \leq M/\underline{\kappa}^{1/2}$, where $\underline{\kappa} > 0$ is from Assumption 4. Moreover,

$$\|z - z'\|_2^2 = (y - y')^\top A(t)A(t)^\top (y - y') = (y - y')^\top I(t)(y - y') \leq \bar{\kappa}\|y - y'\|_2^2,$$

where $\bar{\kappa} < \infty$ is also from Assumption 4. Thus, since $y \in C_{n^{-1/2}}(t) \cap B(0, M/\underline{\kappa}^{1/2})$ and $y' \in C(t) \cap B(0, M/\underline{\kappa}^{1/2})$,

$$\sup_{t \in \mathcal{T}} d_M\{\Delta_n(t), \Delta(t)\} \leq \sqrt{\bar{\kappa}} \sup_{t \in \mathcal{T}} d_{M/\underline{\kappa}^{1/2}}\{C_{n^{-1/2}}(t), C(t)\},$$

which tends to zero by Assumption 7.

To prove the second claim, pick arbitrary $z \in \bar{B}_M$ and $t \in \mathcal{T}$. We have $\|P_{\Delta_n(t)}z\|_2 \leq M$ and $\|P_{\Delta(t)}z\|_2 \leq M$, by definition of the projection since $0 \in \Delta_n(t)$ and $0 \in \Delta(t)$. Thus,

$$\begin{aligned} \left| \|z - P_{\Delta_n(t)}z\|_2^2 - \|z - P_{\Delta(t)}z\|_2^2 \right| &\leq 2M \left| \|z - P_{\Delta_n(t)}z\|_2 - \|z - P_{\Delta(t)}z\|_2 \right| \\ &\leq 2d_M\{\Delta_n(t), \Delta(t)\} \\ &\leq 2M \sup_{t \in \mathcal{T}} d_M\{\Delta_n(t), \Delta(t)\}, \end{aligned}$$

which tends to zero by the first claim.

For the third claim, observe that

$$\|P_{\Delta_n(t)}z\|_2^2 = \|P_{\Delta(t)}z - z + z\|_2^2 = \|z - P_{\Delta(t)}z\|_2^2 + 2z^\top\{z - P_{\Delta(t)}z\} + \|z\|_2^2.$$

so that

$$\|P_{\Delta_n(t)}z\|_2^2 - \|P_{\Delta(t)}z\|_2^2 = \|z - P_{\Delta(t)}z\|_2^2 - \|z - P_{\Delta_n(t)}z\|_2^2 + 2z^\top\{z - P_{\Delta_n(t)}z - (z - P_{\Delta(t)}z)\}.$$

By triangle and Cauchy–Schwarz inequalities, the absolute value of the last display is less than

$$\left| \|z - P_{\Delta(t)}z\|_2^2 - \|z - P_{\Delta_n(t)}z\|_2^2 \right| + 2\|z\|_2 \left| \|z - P_{\Delta_n(t)}z\|_2 - \|z - P_{\Delta(t)}z\|_2 \right|.$$

After taking the supremum over $z \in \bar{B}_M$ and $t \in \mathcal{T}$, the first term tends to zero by the second claim, and it was argued in the proof of that claim that the second term also tends to zero. $\square$

Proof of Theorem 3.10. Since $K(t)$ is the intersection of the closed convex cone $\Delta(t)$ with the closed linear subspace $\{\Delta^0(t)\}^\perp$, it is a closed convex cone contained in $\{\Delta^0(t)\}^\perp$.

For the direct sum, take any $\delta \in \Delta(t)$ and write $\delta = \delta_0 + \kappa$ with $\delta_0 = P_{\Delta^0(t)}\delta \in \Delta^0(t)$ and $\kappa = P_{\{\Delta^0(t)\}^\perp}\delta$. Since $\Delta(t)$ is a convex cone containing the subspace $\Delta^0(t)$, it is closed under addition of its elements, and $-\delta_0 \in \Delta^0(t) \subseteq \Delta(t)$, so $\kappa = \delta + (-\delta_0) \in \Delta(t)$. Thus $\kappa \in K(t)$, giving $\Delta(t) \subseteq \Delta^0(t) + K(t)$. The reverse inclusion follows because $\Delta^0(t) \subseteq \Delta(t)$ and $K(t) \subseteq \Delta(t)$. Orthogonality and uniqueness hold by construction, so $\Delta(t) = \Delta^0(t) \oplus K(t)$.

For the projection identity, write $z = z_0 + z_\perp$ with $z_0 = P_{\Delta^0(t)}z$ and $z_\perp = P_{\{\Delta^0(t)\}^\perp}z$. Any $\delta \in \Delta(t)$ decomposes as $\delta_0 + \kappa$ with $\delta_0 \in \Delta^0(t)$, $\kappa \in K(t)$, $\delta_0 \perp \kappa$, so $\|z - \delta\|_2^2 = \|z_0 - \delta_0\|_2^2 + \|z_\perp - \kappa\|_2^2$. The two terms are minimized independently: the first at $\delta_0 = z_0$ (since $\Delta^0(t)$ is a subspace) and the second at $\kappa = P_{K(t)}z_\perp$, giving $P_{\Delta(t)}z = z_0 + P_{K(t)}z_\perp$. By the Pythagorean theorem, $\|P_{\Delta(t)}z\|_2^2 - \|P_{\Delta^0(t)}z\|_2^2 = \|P_{K(t)}z_\perp\|_2^2$. Finally, since $K(t) \subseteq \{\Delta^0(t)\}^\perp$, for any $z = z_0 + z_\perp$ the component $z_0 \in \Delta^0(t)$ is irrelevant to $P_{K(t)}$, so $P_{K(t)}z_\perp = P_{K(t)}z$, giving the stated identity. Substituting $z = Z(t)$ gives $X(t) - X^0(t) = \|P_{K(t)}Z(t)\|_2^2$ for each t , and taking the supremum yields the result. $\square$

Proof of Lemma 3.4. For an arbitrary $c_2 > 0$ and $c_1 = \underline{\kappa}/2$ pick a c_3 small enough that (10) holds. Let us denote $\bar{B} = \bar{B}_{c_3}(\xi_0; t) = \{\xi \in \Xi(t) : \|\xi - \xi_0\|_2 \leq c_3\}$ for simplicity. Then for all large enough n , with outer probability at least $1 - c_2$, for $\xi \in \bar{B}$,

$$\begin{aligned} \ell_n(\xi, t) - \ell_n(\xi_0, t) &\leq (\xi - \xi_0)^\top n^{1/2} A(t) Z_n(t) - \frac{n\underline{\kappa}}{4} \|\xi - \xi_0\|_2^2 \\ &\leq n^{1/2} \bar{\kappa}^{1/2} \|\xi - \xi_0\|_2 \|Z_n\|_{\mathcal{T}} - \frac{n\underline{\kappa}}{4} \|\xi - \xi_0\|_2^2. \end{aligned}$$

Here, $\underline{\kappa}$ and $\bar{\kappa}$ are from Assumption 4. Let $T_n(\|\xi - \xi_0\|_2)$ denote the quadratic in the last right-hand side. For any $C_1 > 0$, note $\|\hat{\xi}_n(t) - \xi_0\|_2 > \frac{C_1}{\sqrt{n}}$ can only happen if at least one of the following holds: (i) $\hat{\xi}_n(t) \notin \bar{B}$; (ii) the upper bound in the last display fails for some $\xi \in \bar{B}$; or (iii) (i) and (ii) do not hold, in which case taking $\xi = \hat{\xi}_n(t)$ in the upper bound and using that, by definition, $\ell_n(\hat{\xi}_n(t), t) \geq \ell_n(\xi_0, t) - O_{p^*}(1)$ yields $C_1/\sqrt{n} < \|\hat{\xi}_n(t) - \xi_0\|_2 \leq c_3$ and

$$-O_{p^*}(1) \leq \ell_n(\hat{\xi}_n(t), t) - \ell_n(\xi_0, t) \leq T_n(\|\hat{\xi}_n(t) - \xi_0\|_2) \leq \sup_{\|\xi - \xi_0\|_2 > C_1/\sqrt{n}} T_n(\|\xi - \xi_0\|_2).$$

Now (i) has outer probability tending to zero by Assumption 3 and (ii) has outer probability at most c_2 for all large enough n . Thus, it suffices to show that the outer probability of (iii) can be made arbitrarily small asymptotically by choosing C_1 large enough.

To show this, note, for every $C_2 > 0$,

$$\begin{aligned} \mathbf{P}^* \left\{ \sup_{\|\xi - \xi_0\|_2 > C_1/\sqrt{n}} T_n(\|\xi - \xi_0\|_2) \geq -O_{p^*}(1) \right\} &\leq \mathbf{P}^* \left\{ \sup_{\|\xi - \xi_0\|_2 > C_1/\sqrt{n}} T_n(\|\xi - \xi_0\|_2) \geq -C_2/2 \right\} \\ &\quad + \mathbf{P}^* \{O_{p^*}(1) \geq C_2/2\}, \end{aligned}$$

and the last term can be made arbitrarily small asymptotically by choosing C_2 large enough. To control the other term, by the quadratic formula, $T_n(\|\xi - \xi_0\|_2) \geq -C_2/2$ can only hold if

$$\sqrt{n}\|\xi - \xi_0\|_2 \leq \frac{\bar{\kappa}^{1/2}\|Z_n\|_{\mathcal{T}} + \sqrt{\bar{\kappa}\|Z_n\|_{\mathcal{T}}^2 + C_2\bar{\kappa}/2}}{\bar{\kappa}/2}.$$

Thus, $\sup_{\|\xi - \xi_0\|_2 > C_1/\sqrt{n}} nT_n(\|\xi - \xi_0\|_2) \geq -C_2/2$ can only happen if the right-hand side in the last display is greater than C_1 . But that right-hand-side is asymptotically tight by the continuous mapping theorem [van der Vaart and Wellner, 2023, Theorem 1.3.6] since $\|Z_n\|_{\mathcal{T}}$ is by Assumption 5 [van der Vaart and Wellner, 2023, Definition 1.3.7 and Lemma 1.3.8]. Thus, we can pick C_1 large enough to make the outer probability of that event arbitrarily small asymptotically. This completes the proof. $\square$

Lemma A.2. *Under Assumptions 2 * and 5 * , every $\hat{\delta}_n(t)$ such that $\sup_{t \in \mathcal{T}_{ns}} \|\hat{\delta}_n(t)\|_2 = o_{\mathbf{P}^*}(\sqrt{n})$ and $Q_n(\hat{\delta}_n(t), t) + 2\|\hat{\delta}_n(t)\|_2^2 \tilde{R}_n(\hat{\delta}_n(t), t) \geq \sup_{\delta \in \Delta_n(t)} \{Q_n(\delta, t) + 2\|\delta\|_2^2 \tilde{R}_n(\delta, t)\} - O_{\mathbf{P}^*}(1)$ satisfies*

$$\sup_{t \in \mathcal{T}_{ns}} \|\hat{\delta}_n(t)\|_2 = O_{\mathbf{P}^*}(1).$$

Proof of Lemma A.2. Fix an arbitrary $c_2 > 0$ and choose $c_1 \in (0, 1/2)$ (e.g., $c_1 = 1/4$). Pick a c_3 small enough that Assumption 2 * holds. Let us denote $\tilde{B} = \tilde{B}(n, t, c_3)$ for simplicity. Then for all large enough n , with outer probability at least $1 - c_2$, for $\delta \in \tilde{B}$,

$$\begin{aligned} Q_n(\delta, t) + 2\|\delta\|_2^2 \tilde{R}_n(\delta, t) &\leq Q_n(\delta, t) + 2c_1\|\delta\|_2^2 \\ &\leq 2\|\delta\|_2\|Z_n\|_{\mathcal{T}_{ns}} - (1 - 2c_1)\|\delta\|_2^2. \end{aligned}$$

Let $T_n(\|\delta\|_2)$ denote the quadratic in the last right-hand side. For any $C_1 > 0$, note $\sup_{t \in \mathcal{T}_{ns}} \|\hat{\delta}_n(t)\|_2 > C_1$ can only happen if either (i) there exists a $t \in \mathcal{T}_{ns}$ such that $\hat{\delta}_n(t) \notin \tilde{B}$; (ii) the upper bound in the last display does not hold on $\tilde{B}$; or (iii) the bound holds, but for

some $t \in \mathcal{T}_{ns}$ there exists a δ such that $C_1 < \|\delta\|_2 \leq \sqrt{n}c_3$ and

$$-O_{p^*}(1) \leq Q_n(\delta, t) + 2\|\delta\|_2^2 \tilde{R}_n(\delta, t) \leq T_n(\|\delta\|_2) \leq \sup_{\|\delta\|_2 > C_1} T_n(\|\delta\|_2).$$

The last claim uses the definition of $\hat{\delta}_n(t)$ and the upper bound in the preceding display. Now (i) has outer probability tending to zero by the assumption that $\sup_{t \in \mathcal{T}_{ns}} \|\hat{\delta}_n(t)\|_2 = o_{P^*}(\sqrt{n})$ and (ii) has outer probability at most c_2 for all large enough n . Thus, it suffices to show that the outer probability of (iii) can be made arbitrarily small asymptotically by choosing C_1 large enough.

To show this, note, for every $C_2 > 0$,

$$\begin{aligned} P^* \left\{ \sup_{\|\delta\|_2 > C_1} T_n(\|\delta\|_2) \geq -O_{p^*}(1) \right\} &\leq P^* \left\{ \sup_{\|\delta\|_2 > C_1} T_n(\|\delta\|_2) \geq -C_2/2 \right\} \\ &\quad + P^* \{O_{p^*}(1) \geq C_2/2\}, \end{aligned}$$

and the last term can be made arbitrarily small asymptotically by choosing C_2 large enough. To control the other term, by the quadratic formula, $T_n(\|\delta\|_2) \geq -C_2/2$ can only hold if

$$\|\delta\|_2 \leq \|Z_n\|_{\mathcal{T}_{ns}} + \sqrt{\|Z_n\|_{\mathcal{T}_{ns}}^2 + 2c_1 + C_2/2}.$$

Thus, $\sup_{\|\delta\|_2 > C_1} T_n(\|\delta\|_2) \geq -C_2/2$ can only happen if the right-hand side in the last display is greater than C_1 . But that right-hand-side is asymptotically tight since $\|Z_n\|_{\mathcal{T}_{ns}} = O_{P^*}(1)$ by Assumption 5*. Thus, we can pick C_1 large enough to make the outer probability of that event arbitrarily small asymptotically. This completes the proof. $\square$

Lemma A.3. *Under Assumptions 2* and 5*, if $\sup_{t \in \mathcal{T}_{ns}} \|\hat{\delta}_n(t)\|_2 = o_{P^*}(\sqrt{n})$, then $\sup_{t \in \mathcal{T}_{ns}} |X_n(t) - \tilde{X}_n(t)| = o_{P^*}(1)$.*

Proof of Lemma A.3. Pick $\hat{\delta}_n(t)$ such that $Q_n(\hat{\delta}_n(t), t) + 2\|\hat{\delta}_n(t)\|_2^2 \tilde{R}_n(\hat{\delta}_n(t), t) \geq \sup_{\delta \in \Delta_n(t)} \{Q_n(\delta, t) + 2\|\delta\|_2^2 \tilde{R}_n(\delta, t)\} - 1/n$. Assuming $\sup_{t \in \mathcal{T}_{ns}} \|\hat{\delta}_n(t)\|_2 = o_{P^*}(\sqrt{n})$, Lemma A.2 gives $\sup_{t \in \mathcal{T}_{ns}} \|\hat{\delta}_n(t)\|_2 = O_{P^*}(1)$. Pick a $\tilde{\delta}_n(t)$ such that $Q_n(\tilde{\delta}_n(t), t) \geq \sup_{\delta \in \Delta_n(t)} Q_n(\delta, t) - 1/n$. Since $0 \in \Delta_n(t)$ and $Q_n(0, t) = 0$, the supremum is non-negative, so $Q_n(\tilde{\delta}_n(t), t) \geq -1/n$. But for any δ with $\|\delta\|_2 > 2\|Z_n(t)\|_2$, by Cauchy–Schwarz,

$$Q_n(\delta, t) \leq 2\|\delta\|_2 \|Z_n(t)\|_2 - \|\delta\|_2^2 = \|\delta\|_2 (2\|Z_n(t)\|_2 - \|\delta\|_2) < 0. \quad (\text{A.2})$$

Thus, if $\|Z_n\|_{\mathcal{T}_{ns}} \leq C_1$, any $\|\delta\|_2 > 3C_1$ leads to a value less than $-3C_1$, which is less than

$-1/n$ for all large enough n . Thus, for such n , $\sup_{t \in \mathcal{T}_{ns}} \|\tilde{\delta}_n(t)\|_2 \leq 3C_1$ whenever $\|Z_n\|_{\mathcal{T}_{ns}} \leq C_1$. Since $\|Z_n\|_{\mathcal{T}_{ns}} = O_{\mathbf{P}^*}(1)$ by Assumption 5*, we have $\sup_{t \in \mathcal{T}_{ns}} \|\tilde{\delta}_n(t)\|_2 = O_{\mathbf{P}^*}(1)$.

Next, for any $c_1, C_1 > 0$, by countable sub-additivity of outer probabilities,

$$\begin{aligned} \mathbf{P}^* \left(\sup_{t \in \mathcal{T}_{ns}} |X_n(t) - \tilde{X}_n(t)| > c_1 \right) &\leq \mathbf{P}^* \left(\sup_{t \in \mathcal{T}_{ns}} |X_n(t) - \tilde{X}_n(t)| > c_1, \sup_{t \in \mathcal{T}_{ns}} \|\hat{\delta}_n(t)\|_2 \leq C_1, \sup_{t \in \mathcal{T}_{ns}} \|\tilde{\delta}_n(t)\|_2 \leq C_1 \right) \\ &\quad + \mathbf{P}^* \left(\sup_{t \in \mathcal{T}_{ns}} \|\hat{\delta}_n(t)\|_2 > C_1 \right) + \mathbf{P}^* \left(\sup_{t \in \mathcal{T}_{ns}} \|\tilde{\delta}_n(t)\|_2 > C_1 \right). \end{aligned}$$

The last two terms can be made arbitrarily small by choosing C_1 large enough, so let us deal with the first term. Define $\tilde{\Upsilon}_{C_1, n} = \{(\delta, t) : t \in \mathcal{T}_{ns}, \delta \in \Delta_n(t), \|\delta\|_2 \leq C_1\}$. Then, using that the supremum of the difference is no greater than the difference of suprema, for outcomes where $\sup_{t \in \mathcal{T}_{ns}} \|\hat{\delta}_n(t)\|_2 \leq C_1$ and $\sup_{t \in \mathcal{T}_{ns}} \|\tilde{\delta}_n(t)\|_2 \leq C_1$,

$$\begin{aligned} \sup_{t \in \mathcal{T}_{ns}} |X_n(t) - \tilde{X}_n(t)| &= \sup_{t \in \mathcal{T}_{ns}} \left| \sup_{\delta \in \Delta_n(t)} \{Q_n(\delta, t) + 2\|\delta\|_2^2 \tilde{R}_n(\delta, t)\} - \sup_{\delta \in \Delta_n(t)} Q_n(\delta, t) \right| \\ &\leq 2/n + \sup_{t \in \mathcal{T}_{ns}} \left| Q_n(\hat{\delta}_n(t), t) + 2\|\hat{\delta}_n(t)\|_2^2 \tilde{R}_n(\hat{\delta}_n(t), t) - Q_n(\hat{\delta}_n(t), t) \right| \\ &\leq 4/n + \sup_{t \in \mathcal{T}_{ns}} \left| \sup_{\delta \in \Delta_n(t), \|\delta\|_2 \leq C_1} \{Q_n(\delta, t) + 2\|\delta\|_2^2 \tilde{R}_n(\delta, t)\} - \sup_{\delta \in \Delta_n(t), \|\delta\|_2 \leq C_1} Q_n(\delta, t) \right| \\ &\leq 4/n + \sup_{(\delta, t) \in \tilde{\Upsilon}_{C_1, n}} 2\|\delta\|_2^2 |\tilde{R}_n(\delta, t)|. \end{aligned}$$

By Assumption 2*, for any $c_1 > 0$, the outer probability that $\sup_{(\delta, t) \in \tilde{\Upsilon}_{C_1, n}} |\tilde{R}_n(\delta, t)| > c_1$ can be made arbitrarily small for all large enough n by choosing c_3 small enough, because for any fixed C_1 , $C_1 \leq \sqrt{n}c_3$ for all large enough n , so the supremum is taken over a subset of $\tilde{B}(n, t, c_3)$. For outcomes where $\sup_{t \in \mathcal{T}_{ns}} \|\hat{\delta}_n(t)\|_2 \leq C_1$ and $\sup_{t \in \mathcal{T}_{ns}} \|\tilde{\delta}_n(t)\|_2 \leq C_1$, we have $\|\delta\|_2 \leq C_1$ for $(\delta, t) \in \tilde{\Upsilon}_{C_1, n}$, so the last term is bounded by $2C_1^2 \sup_{(\delta, t) \in \tilde{\Upsilon}_{C_1, n}} |\tilde{R}_n(\delta, t)|$ and hence is $o_{\mathbf{P}^*}(1)$, which finishes the proof. $\square$

Lemma A.4. *Under Assumptions 4*, 5*, 6, and 7, $\sup_{t \in \mathcal{T}_\epsilon} |\tilde{X}_n(t) - \check{X}_n(t)| = o_{\mathbf{P}^*}(1)$.*

Proof of Lemma A.4. By the same arguments as in the proof of Lemma A.1, replacing $\mathcal{T}$ with $\mathcal{T}_\epsilon$ and using Assumption 4* in place of Assumption 4, we have $\lim_{n \rightarrow \infty} \sup_{t \in \mathcal{T}_\epsilon} d(\Delta_n(t), \Delta(t)) = 0$. Consequently, for any $C_1 > 0$,

$$\lim_{n \rightarrow \infty} \sup_{t \in \mathcal{T}_\epsilon, \|z\|_2 \leq C_1} \left| \|P_{\Delta_n(t)} z\|_2^2 - \|P_{\Delta(t)} z\|_2^2 \right| = 0.$$

As in the proof of Lemma 3.6, we have $\tilde{X}_n(t) = \|Z_n(t)\|_2^2 - \|P_{\Delta_n(t)}Z_n(t)\|_2^2$ and $\check{X}_n(t) = \|Z_n(t)\|_2^2 - \|P_{\Delta(t)}Z_n(t)\|_2^2$. Thus, for outcomes where $\|Z_n\|_{\mathcal{T}_\epsilon} \leq C_1$,

$$\begin{aligned} \sup_{t \in \mathcal{T}_\epsilon} |\tilde{X}_n(t) - \check{X}_n(t)| &= \sup_{t \in \mathcal{T}_\epsilon} \left| \|P_{\Delta(t)}Z_n(t)\|_2^2 - \|P_{\Delta_n(t)}Z_n(t)\|_2^2 \right| \\ &\leq \sup_{t \in \mathcal{T}_\epsilon, \|z\|_2 \leq C_1} \left| \|P_{\Delta_n(t)}z\|_2^2 - \|P_{\Delta(t)}z\|_2^2 \right|, \end{aligned}$$

which tends to zero. Since $\|Z_n\|_{\mathcal{T}_\epsilon} = O_{\mathbf{P}^*}(1)$ by Assumption 5^{*} (noting $\mathcal{T}_\epsilon \subseteq \mathcal{T}_{ns}$), the outer probability of $\|Z_n\|_{\mathcal{T}_\epsilon} > C_1$ can be made arbitrarily small by choosing C_1 large enough, completing the proof. $\square$

Lemma A.5. *Under Assumptions 5^{*}, 6, and 7, $\check{X}_n \xrightarrow{\mathcal{L}} X$ in $F(\mathcal{T}_{ns})$, where $X(t) = \sup_{\delta \in \Delta(t)} (\delta^\top Z(t) - \|\delta\|_2^2/2)$.*

Proof of Lemma A.5. Define the map $g : F^p(\mathcal{T}_{ns}) \rightarrow F(\mathcal{T}_{ns})$ by $g(z)(t) = \sup_{\delta \in \Delta(t)} (\delta^\top z(t) - \|\delta\|_2^2/2)$. For any $z_1, z_2 \in F^p(\mathcal{T}_{ns})$, let $\hat{\delta}_i(t)$ be the projection of $z_i(t)$ onto $\Delta(t)$ for $i = 1, 2$. Since $0 \in \Delta(t)$, $\|\hat{\delta}_i(t)\|_2 \leq \|z_i(t)\|_2 \leq \|z_i\|_{\mathcal{T}_{ns}}$. Thus,

$$\begin{aligned} |g(z_1)(t) - g(z_2)(t)| &\leq \sup_{\delta \in \Delta(t)} |\delta^\top (z_1(t) - z_2(t))| \\ &\leq \max(\|\hat{\delta}_1(t)\|_2, \|\hat{\delta}_2(t)\|_2) \|z_1(t) - z_2(t)\|_2 \\ &\leq \max(\|z_1\|_{\mathcal{T}_{ns}}, \|z_2\|_{\mathcal{T}_{ns}}) \|z_1 - z_2\|_{\mathcal{T}_{ns}}. \end{aligned}$$

Taking the supremum over $t \in \mathcal{T}_{ns}$, we see that g is continuous on bounded sets in $F^p(\mathcal{T}_{ns})$. Since Z is tight, it concentrates on a separable subspace, and the continuous mapping theorem applies. Thus, $\check{X}_n = g(Z_n) \xrightarrow{\mathcal{L}} g(Z) = X$ in $F(\mathcal{T}_{ns})$. $\square$

Proof of Corollary 3.12. For a fixed n , define

$$g_n(\xi, t) = 2\{\ell_n(\xi, t) - \ell_n(\xi_0, t)\}, \quad X_n(t) = \sup_{\xi \in \Xi(t)} g_n(\xi, t).$$

We first show that $t \mapsto X_n(t)$ is lower semicontinuous on $\mathcal{T}$. Let $t_m \rightarrow t$ in $\mathcal{T}$ and fix $\eta > 0$. Choose $\xi \in \Xi(t)$ such that $g_n(\xi, t) \geq X_n(t) - \eta$. By inner semicontinuity of $\Xi(\cdot)$, there exist $\xi_m \in \Xi(t_m)$ such that $\xi_m \rightarrow \xi$. Since ℓ_n is lower semicontinuous, $\liminf_{m \rightarrow \infty} \ell_n(\xi_m, t_m) \geq \ell_n(\xi, t)$, and therefore

$$\liminf_{m \rightarrow \infty} X_n(t_m) \geq \liminf_{m \rightarrow \infty} g_n(\xi_m, t_m) \geq g_n(\xi, t) \geq X_n(t) - \eta.$$

Letting $\eta \downarrow 0$ proves lower semicontinuity of X_n .

By Assumption 8, $\mathcal{T}_{ns}$ is dense in $\mathcal{T}$. For any lower semicontinuous function f on a metric space, density implies $\sup_{t \in \mathcal{T}} f(t) = \sup_{t \in \mathcal{T}_{ns}} f(t)$, because for any $t \in \mathcal{T}$ one can pick $t_m \in \mathcal{T}_{ns}$ with $t_m \rightarrow t$ and then $f(t) \leq \liminf_m f(t_m) \leq \sup_{\mathcal{T}_{ns}} f$. Applying this with $f = X_n$ yields

$$\Lambda_n = \sup_{t \in \mathcal{T}} X_n(t) = \sup_{t \in \mathcal{T}_{ns}} X_n(t) = \Lambda_n^{(ns)}.$$

Next, define the extension X as in the corollary: extend Z to be almost surely continuous on $\mathcal{T}$, and for $t \in \mathcal{T}_s$ define $\Delta(t)$ as the inner limit of $\Delta(t')$ as $t' \rightarrow t$. Then $X(t)$ is given by

$$X(t) = \sup_{\delta \in \Delta(t)} \{2\delta^\top Z(t) - \|\delta\|_2^2\}.$$

The same argument as above (with ξ replaced by δ and using inner semicontinuity of $\Delta(\cdot)$ and continuity of Z) shows that X is lower semicontinuous on $\mathcal{T}$. Hence, again by density of $\mathcal{T}_{ns}$,

$$\Lambda = \sup_{t \in \mathcal{T}} X(t) = \sup_{t \in \mathcal{T}_{ns}} X(t) = \Lambda^{(ns)}.$$

Finally, Theorem 3.11 gives $\Lambda_n^{(ns)} \xrightarrow{\mathcal{L}} \Lambda^{(ns)}$. Since $\Lambda_n = \Lambda_n^{(ns)}$ and $\Lambda = \Lambda^{(ns)}$, we conclude $\Lambda_n \xrightarrow{\mathcal{L}} \Lambda$. $\square$

Proof of Proposition 4.1. For a multi-index $k = (k_1, \dots, k_p)$ of non-negative integers, write $|k| = k_1 + \dots + k_p$, $k! = k_1! \dots k_p!$, $\partial_\xi^k = \partial^{k_1} / \partial \xi_1^{k_1} \dots \partial \xi_p^{k_p}$, and $(\xi - \xi_0)^k = (\xi_1 - \xi_{0,1})^{k_1} \dots (\xi_p - \xi_{0,p})^{k_p}$. Condition (iii) of Proposition 4.1 means formally that there exists a measurable $M : \mathcal{Y} \rightarrow \mathbb{R}$ with $\mathbf{E}\{M(Y_1)\} < \infty$ and, for some $c_1 > 0$,

$$\sup_{\theta \in \Theta : \|\xi - \xi_0\|_2 \leq c_1} |\partial_\xi^k \log f(y; \theta)| \leq M(y)$$

for all k with $|k| = 3$ and ν -almost every $y \in \mathcal{Y}$. Recall (3) and note, by Taylor's theorem, for any fixed $t \in \mathcal{T}$, there exists some $\tilde{\xi}$ between ξ_0 and ξ such that

$$n\|I(t)^{1/2}(\xi - \xi_0)\|_2^2 R_n(t, \xi) = \frac{1}{3!} \sum_{k: |k|=3} \frac{3!}{k!} \partial_\xi^k \ell_n(\tilde{\xi}, t) (\xi - \xi_0)^k + \frac{n}{2} (\xi - \xi_0)^\top \{I(t) - \hat{I}_n(t)\} (\xi - \xi_0).$$

By the triangle inequality and condition (iii) of Proposition 4.1, the absolute value of the

first term is upper bounded by

$$\frac{1}{3!} \sum_{k:|k|=3} \frac{3!}{k!} \sum_{i=1}^n M(Y_i) |\xi - \xi_0|^k = \frac{1}{3!} \sum_{i=1}^n M(Y_i) (|\xi - \xi_0|^\top \mathbf{1}_p)^3 \leq \frac{1}{3!} p^{3/2} \|\xi - \xi_0\|_2^3 \sum_{i=1}^n M(Y_i).$$

The absolute value of the second term is bounded by $(n/2) \|\xi - \xi_0\|_2^2 \|I - \hat{I}_n\|_{\mathcal{T}}$. Combining the two bounds and using $\|I(t)^{1/2}(\xi - \xi_0)\|_2^2 \geq \underline{\kappa} \|\xi - \xi_0\|_2^2$ from Assumption 4 gives

$$|R_n(t, \xi)| \leq \underline{\kappa}^{-1} \left\{ \frac{1}{3!} p^{3/2} \|\xi - \xi_0\|_2 n^{-1} \sum_{i=1}^n M(Y_i) + \frac{1}{2} \|I - \hat{I}_n\|_{\mathcal{T}} \right\}.$$

The first term is asymptotically uniformly equicontinuous almost surely, and hence in outer probability, since $n^{-1} \sum_{i=1}^n M(Y_i) \rightarrow \mathbf{E}\{M(Y_1)\}$ almost surely by the strong law of large numbers. The second term does not depend on ξ or t and is $o_{p^*}(1)$ by condition (ii), so the sum is asymptotically uniformly equicontinuous in outer probability, completing the proof. $\square$

Proof of Theorem 5.4. With a slight abuse of notation, write $\ell_n(\xi, t)$ for the linkage log likelihood with t and ξ as in (34), so that $\xi_0 = (0, 0)$ corresponds to the null. Let $\varepsilon = (\varepsilon_1, \varepsilon_2)$ be such that $\xi = (\varepsilon_1^2, \varepsilon_2^2)$ with $\varepsilon_1, \varepsilon_2 \geq 0$, and write $\ell_n(t, \varepsilon)$ when we view the same log-likelihood as a function of ε .

Following Rotnitzky et al. [2000], expand $\ell_n(t, \varepsilon)$ around $\varepsilon = 0$ up to order 4:

$$\ell_n(t, \varepsilon) = \ell_n(t, 0) + \sum_{k: 1 \leq |k| \leq 4} \ell_n^{(k)}(t, 0) \frac{\varepsilon^k}{k!} + \bar{R}_n(t; \varepsilon), \quad (\text{A.3})$$

where $\ell_n^{(k)}(t, \varepsilon) = \partial^k \ell_n(t, \varepsilon) / \partial \varepsilon^k$, $k = (k_1, k_2)$, and $\varepsilon^k = \varepsilon_1^{k_1} \varepsilon_2^{k_2}$. Moreover, $\ell_n^{(k)}(t, 0) = \sum_{i=1}^n \psi_k(X_i; t)$, where

$$\psi_k(x; t) = \frac{\partial^k \log f_{(t, \varepsilon)}(x)}{\partial \varepsilon^k} = \frac{\partial^k \log \mathbf{P}_{\mathbf{p}, \gamma}(v(\tau) | \Phi)}{\partial \varepsilon^k}. \quad (\text{A.4})$$

Using results from Hössjer [2005a] and repeated differentiation with respect to ε at $\varepsilon = 0$, one obtains in particular that the only nonzero contributions up to order 4 come from the even derivatives, with

$$\psi_{20}(x; t) = 2S_1(v(\tau)), \quad \psi_{02}(x; t) = 2S_2(v(\tau)), \quad \psi_{11}(x; t) = 0,$$

where $S = (S_1, S_2)$ is as in (36), and

$$\psi_{40}(x; t) = -3I_{11}(t, t) + r_{40}(x; t), \quad \psi_{22}(x; t) = -I_{12}(t, t) + r_{22}(x; t), \quad \psi_{04}(x; t) = -3I_{22}(t, t) + r_{04}(x; t),$$

with $I(t, t) = E_0\{S(v(\tau))^T S(v(\tau))\}$ as in (38) and $E_0\{r_k(X; t)\} = 0$ for $k \in \{(4, 0), (2, 2), (0, 4)\}$.

Substituting $\xi = (\varepsilon_1^2, \varepsilon_2^2)$ into (A.3) and collecting terms, we arrive at the quadratic expansion in the notation of Section 2:

$$\ell_n(\xi, t) - \ell_n(\xi_0) = \sqrt{n}(\xi - \xi_0)^T U_n(t) - \frac{n}{2}(\xi - \xi_0)^T I(t)(\xi - \xi_0) + n\|I(t)^{1/2}(\xi - \xi_0)\|_2^2 R_n(\xi, t), \quad (\text{A.5})$$

where

$$U_n(t) = n^{-1/2} \nabla_\xi \ell_n(\xi_0, t), \quad \nabla_\xi \ell(\xi_0, t) = S(v(\tau); K) \text{ as in (36),}$$

and $I(t) = I(t, t)$ from (38). This identifies the $U_n(t)$ and $I(t)$ appearing in (3), and yields $Z_n(t) = A(t)^{-1} U_n(t)$ as in Section 2.

It remains to check that the remainder can be written in the form $n\|I(t)^{1/2}(\xi - \xi_0)\|_2^2 R_n(\xi, t)$ with R_n satisfying Assumption 2. By Taylor's theorem, the remainder $\bar{R}_n(t; \varepsilon)$ in (A.3) can be expressed using derivatives of order 5 evaluated at points between 0 and ε . Under a regularity condition analogous to the ones used in Section 3 (existence of an integrable envelope that bounds the order-5 derivatives uniformly over $t \in \mathcal{T}$ and ε in a neighborhood of 0), we obtain the uniform bound

$$|\bar{R}_n(t; \varepsilon)| \leq \left(\sum_{i=1}^n M(X_i) \right) \|\varepsilon\|_2^5 \quad \text{for all } t \in \mathcal{T} \text{ and } \|\varepsilon\|_2 \text{ small,}$$

for some M with $E_0\{M(X_1)\} < \infty$. Since $\|\varepsilon\|_2^4 = \|\xi - \xi_0\|_2^2$ and, by Assumption 4, $\|I(t)^{1/2}(\xi - \xi_0)\|_2^2 \geq \underline{\kappa} \|\xi - \xi_0\|_2^2$, this yields

$$\sup_{t \in \mathcal{T}} \sup_{\xi \in \bar{B}_{c_3}(\xi_0; t)} \frac{|\bar{R}_n(t; \varepsilon)|}{n\|I(t)^{1/2}(\xi - \xi_0)\|_2^2} \leq \underline{\kappa}^{-1} \left(\frac{1}{n} \sum_{i=1}^n M(X_i) \right) c_3^{1/2},$$

and the right-hand side is $O_p(1)$ by the law of large numbers. Choosing c_3 small enough gives Assumption 2, completing the proof. □

A.2 Details for Section 5.2

Derivation of $P_{p,\gamma}(v(\tau)|\Phi)$ in (31). Under H_1 , the phenotype vector $\Phi = (\Phi_k, k \in \mathcal{P})$ of pedigree $\mathcal{P}$ depends on the inheritance vector $v(\tau)$ of the family at the disease locus $\tau \in [0, T]$, and marker data $Y = \{v(s); 0 \leq s \leq T\}$ of the family is conditionally independent of Φ given $v(\tau)$. Since the null distribution of $v(\tau)$ is uniform (32), by Bayes' rule,

$$P_{p,\gamma}(v(\tau)|\Phi) \propto P_{p,\gamma}(\Phi|v(\tau)). \quad (\text{A.6})$$

Let F be the number of founders, $N = F + m/2$ the total number of pedigree members, and $G = (G_1, \dots, G_N)$ the genotypes at τ , where $G_k = (a_{2k-1}, a_{2k})$ contains the two alleles of individual k . For a biallelic gene with $a_k \in \{0, 1\}$ and probability $P(1) = p$ of the disease allele, the founder alleles $a = (a_1, \dots, a_{2F})$ have distribution $P_p(a) = p^{|a|}q^{2F-|a|}$, with $|a| = \sum a_k$ and $q = 1 - p$. Since genotypes of nonfounders are determined by a and $v = v(\tau)$ via $G = G(a, v)$, conditioning on a gives

$$P_{p,\gamma}(\Phi|v) = \sum_a p^{|a|}q^{2F-|a|} P_\gamma(\Phi|G(a, v)). \quad (\text{A.7})$$

For a monogenic binary phenotype ($\Phi_k \in \{0, 1, ?\}$) with conditionally independent phenotypes given genotypes (a typical assumption of a monogenic disease),

$$P_\gamma(\Phi|G) = \prod_{k=1}^N P_\gamma(\Phi_k|G_k), \quad P_\gamma(\Phi_k|G_k) = \gamma_{|G_k|}^{\{\Phi_k=1\}}(1 - \gamma_{|G_k|}^{\{\Phi_k=0\}}), \quad (\text{A.8})$$

and $|G_k| = a_{2k-1} + a_{2k}$.

Phenotype-based weights for pairs of individuals, for score function (36). It can be shown [Hössjer, 2005a] that the weight assigned to a pair (k, l) of individuals, for the score vector components S_1 and S_2 in (36), is given by

$$\omega_{kl} = \frac{\partial^2 P(\Phi|G) / \partial \gamma_{|G_k|} \partial \gamma_{|G_l|} \big|_{\gamma=(K,K,K)}}{P(\Phi)}, \quad (\text{A.9})$$

with $P(\Phi|G)$ the conditional probability defined in (A.8).

Reparametrization of penetrance parameters. For fixed p , introduce the inner product $(x, y) = q^2 x_0 y_0 + 2qp x_1 y_1 + p^2 x_2 y_2$ on $\mathbb{R}^3$ and the orthonormal basis

$$e_0 = (1, 1, 1), \quad e_1 = \frac{(-2p, q - p, 2q)}{\sqrt{2pq}}, \quad e_2 = (q^{-1} - 1, -1, p^{-1} - 1).$$

Writing $\gamma = K e_0 + \varepsilon_1 e_1 + \varepsilon_2 e_2$ as in (33), one checks that $K = E(\Phi_k)$ is the disease prevalence and $\varepsilon_1^2 + \varepsilon_2^2 = \text{Var}(E(\Phi_k | G_k))$ is the total genetic variance split into additive ($\xi_1 = \varepsilon_1^2$) and dominance ($\xi_2 = \varepsilon_2^2$) components.

Derivation of (43). We generalize Theorem 2 of Hössjer [2005b] from one- to two-dimensional score functions. We introduce the space $\mathcal{A}$ of mappings $\{0, 1\}^m \rightarrow \mathbb{R}$, so that both components of the score function $S = (S_1, S_2)^T$ are elements of $\mathcal{A}$. Endow $\mathcal{A}$ with the scalar product

$$\langle S_1, S_2 \rangle = 2^{-m} \sum_w S_1(w) S_2(w).$$

Given any $w \in \{0, 1\}^m$, let $S_w(u) = (-1)^{w \cdot u}$, where $w \cdot u = \sum_{l=1}^m w_l u_l$ is the vector dot product of w and u . Then $\{S_w\}$ is an orthonormal basis on $\mathcal{A}$ and S can be expanded as

$$S = \sum_w R_S(w) S_w,$$

where $R_S(w) = (R_{S_1}(w), R_{S_2}(w))^T$ and $R_{S_k}(w) = \langle S_k, S_w \rangle$. Notice that $R_S(0) = 0$ since both S_1 and S_2 are standardized to have mean zero, under the null hypothesis of no linkage, i.e. $E_{\xi_0}(S_k) = \langle S_k, S_0 \rangle = 0$. It follows from (38) and the proof of Theorem 2 in Hössjer [2005b] that

$$I(t, t') = \sum_{w \neq 0} R_S(w) R_S(w)^T \exp(-2|w| |\tau' - \tau|), \quad (\text{A.10})$$

where $|w| = \sum_{l=1}^m w_l$ is the number of one-components of w . Combining (A.10) with (22) and (42) we arrive at (43), with

$$\begin{aligned} \kappa_l &= A(t)^{-1} \sum_{w: |w|=l} R_S(w) R_S(w)^T A(t)^{-T} \\ &= B^{-1} \sum_{w: |w|=l} R_{S_0}(w) R_{S_0}(w)^T B^{-T}. \end{aligned} \quad (\text{A.11})$$

where $S_0 = K^2 S = (S_{\text{pairs}} - E_{\xi_0}(S_{\text{pairs}}), S_{\text{g-prs}} - E_{\xi_0}(S_{\text{g-prs}}))^T$ and $B = K^2 A(t)$ are both independent of t . $\square$

MLS score for affected sib pairs. For affected sib pairs (pedigree type 1 in Figure 1), the MLS parametrization uses parameter vector $\theta = (\xi, t)$, with

$$t = \tau, \quad \xi = (z_0, z_1), \quad (\text{A.12})$$

where z_i is the probability that the sib pair shares i alleles IBD at τ , with a simple null hypothesis value $\xi_0 = (0.25, 0.5)$ and parameter space for the structural parameter ξ the possible triangle [Holmans, 1993]:

$$\Xi = \{\xi : z_0 \geq 0, z_1 \leq 0.5, 2z_0 \leq z_1\}. \quad (\text{A.13})$$

An affected sib pair has $m = 4$ meioses and marker data $Y = \{v(s) = (v_1(s), \dots, v_4(s)); 0 \leq s \leq T\}$. Number the four family members so that the parents have numbers 1,2 and the siblings numbers 3,4. The density of marker data Y conditionally on the phenotype vector $\Phi = (?, ?, 1, 1)$ is

$$f(y; \theta) = P_\xi(\text{IBD}(\tau)|\Phi) P(v(\tau)|\text{IBD}(\tau)) P(y|v(\tau)), \quad (\text{A.14})$$

where $P(y|v(\tau)) = \exp(-mT) = \exp(-4T)$ is a constant, $\text{IBD}(\tau)$ is the number of alleles shared IBD by the sib pair at the disease locus τ , $P_\xi(\text{IBD}|\Phi) = z_0^{\{\text{IBD}=0\}} z_1^{\{\text{IBD}=1\}} (1 - z_0 - z_1)^{\{\text{IBD}=2\}}$, and $P(v(\tau)|\text{IBD}(\tau)) = 1/8$ if $\text{IBD}(\tau) = 1$ and $1/4$ otherwise. The per-family score vector is

$$\begin{aligned} \nabla_\xi \log f(y; \xi, t) &= \left(4 \cdot \mathbf{1}_{\{\text{IBD}(\tau)=0\}} - 4 \cdot \mathbf{1}_{\{\text{IBD}(\tau)=2\}}, 2 \cdot \mathbf{1}_{\{\text{IBD}(\tau)=0\}} - 4 \cdot \mathbf{1}_{\{\text{IBD}(\tau)=2\}} \right)^T \\ &= S(v(\tau)), \end{aligned} \quad (\text{A.15})$$

where $S(v(\tau))$ is as in (41) for pedigree type 1, so that the information matrices agree:

$$I(t) = \begin{pmatrix} 8 & 4 \\ 4 & 6 \end{pmatrix}. \quad (\text{A.16})$$

From (A.13), the tangent cone at ξ_0 of $\Xi - \xi_0$ is $C = \{(c_1, c_2) : c_2 \leq 0, 2c_1 \leq c_2\}$, which in matrix form (24) corresponds to $C = C_V$, with

$$V = \begin{pmatrix} -2 & 1 \\ 0 & -1 \end{pmatrix}. \quad (\text{A.17})$$

Inserting (A.16) and (A.17) into $\Delta(t) = A(t)^T C = \Delta$ and the definition of w_2 in (40), and writing $\Delta = C_U$ with rows of U normalized to unit length, one obtains

$$U = \begin{pmatrix} -0.5774 & 0.8165 \\ 0 & -1.0000 \end{pmatrix}, \quad w_2 = 0.0980,$$

in agreement with the affected sib pair entry of Table 2.

Table 1: Matrices $\kappa_l = (\kappa_{l,uv})_{u,v=1}^2$ in (43) for the pedigree types $(\mathcal{P}_j, \Phi_j)$ of Figure 1. The upper triangular square root $A(t)^T$ of $I(t)$ is used in (A.11).

j	κ_1	κ_2	κ_3
1-4	$\begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$	$\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$	$\begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$
5	$\begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$	$\begin{pmatrix} 0.8137 & 0.1976 \\ 0.1976 & 0.0821 \end{pmatrix}$	$\begin{pmatrix} 0.1765 & -0.1872 \\ -0.1872 & 0.1985 \end{pmatrix}$
6	$\begin{pmatrix} 0.1914 & -0.1375 \\ -0.1375 & 0.0988 \end{pmatrix}$	$\begin{pmatrix} 0.5455 & 0.3265 \\ 0.3265 & 0.2601 \end{pmatrix}$	$\begin{pmatrix} 0.2161 & -0.1552 \\ -0.1552 & 0.1115 \end{pmatrix}$
7	$\begin{pmatrix} 0.1356 & -0.1246 \\ -0.1246 & 0.1144 \end{pmatrix}$	$\begin{pmatrix} 0.7034 & 0.2725 \\ 0.2725 & 0.1352 \end{pmatrix}$	$\begin{pmatrix} 0.1525 & -0.1401 \\ -0.1401 & 0.1287 \end{pmatrix}$
j	κ_4	κ_5	κ_6
1-4	$\begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}$	$\begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$	$\begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$
5	$\begin{pmatrix} 0.0098 & -0.0104 \\ -0.0104 & 0.7194 \end{pmatrix}$	$\begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$	$\begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$
6	$\begin{pmatrix} 0.0425 & -0.0306 \\ -0.0306 & 0.5273 \end{pmatrix}$	$\begin{pmatrix} 0.0043 & -0.0031 \\ -0.0031 & 0.0022 \end{pmatrix}$	$\begin{pmatrix} 0.0002 & -0.0002 \\ -0.0002 & 0.0001 \end{pmatrix}$
7	$\begin{pmatrix} 0.0085 & -0.0078 \\ -0.0078 & 0.6217 \end{pmatrix}$	$\begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$	$\begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$

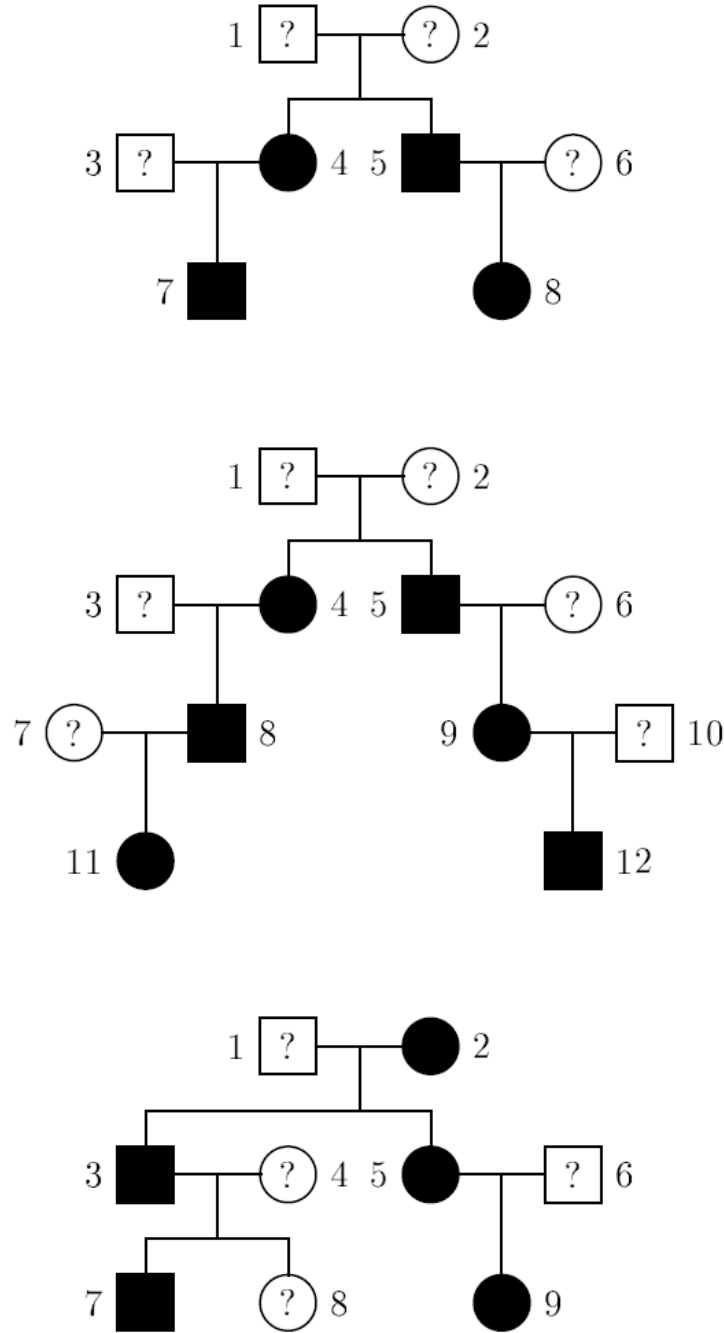


Figure 1: Pedigree types $(\mathcal{P}_j, \Phi_j)$ used in Tables 1-2. For $j = 1, 2, 3, 4$, $\mathcal{P}_j$ consists of two parents with unknown phenotypes and $k + 1$ affected offspring. (In particular, $(\mathcal{P}_1, \Phi_1)$ is an affected sib pair.) $\mathcal{P}_5$ (upper), $\mathcal{P}_6$ (middle) and $\mathcal{P}_7$ (lower) are shown above with individual numbers. Males and females correspond to squares and circles, affected individuals have black and unaffected ones have white symbols. Individuals with unknown phenotypes have question marks.

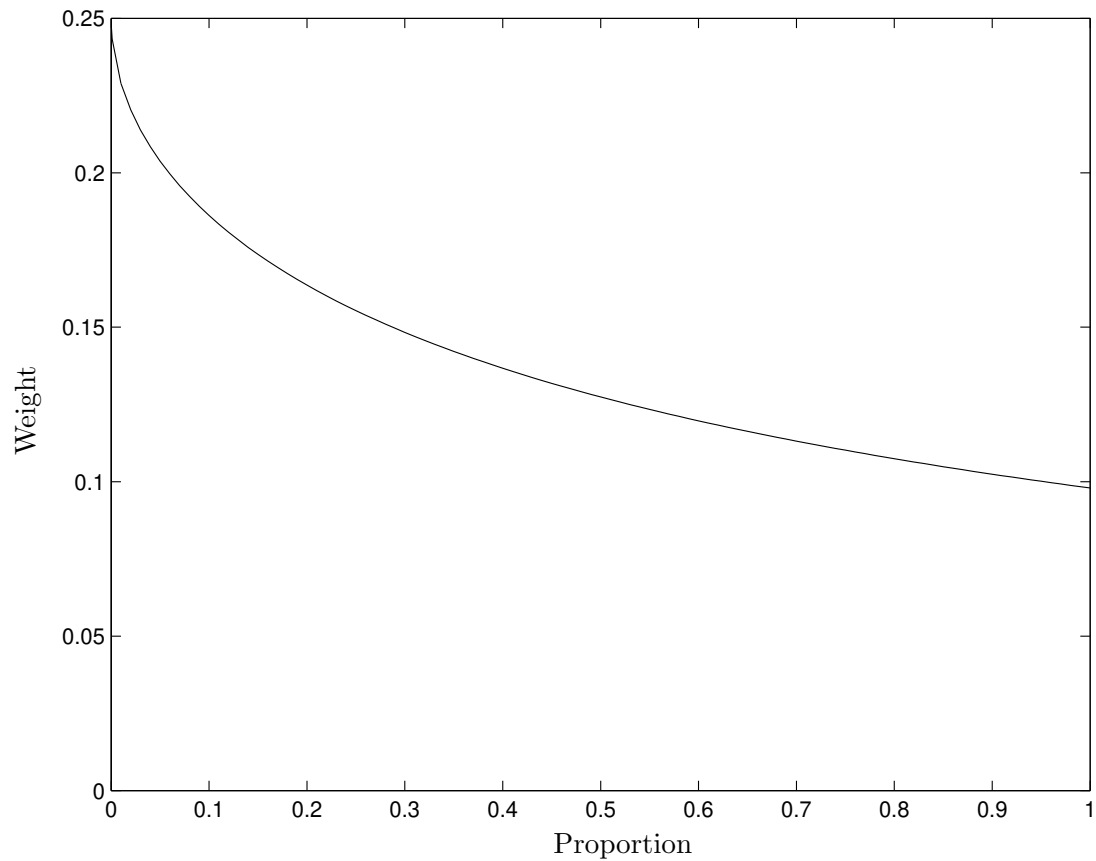


Figure 2: Plot of χ^2_2 -weight w_2 as function of the proportion β of affected sib pairs for a mixture of affected sib and first cousin pairs.

Table 2: Values of Fisher information matrix $I(t)$, boundary region $\Delta = C_U$ and χ^2_2 -weight w_2 (see (40)) for the pedigree types of Figure 1. The upper triangular square root $A(t)^\top$ of $I(t)$ is used for calculating $U(t) = A(t)^{-\top}$. After normalizing the two rows of $U(t)$ to have unit length, the resulting matrix U does not depend on t .

Pedigree type	$I(t)$	U	w_2
$(\mathcal{P}_1, \Phi_1)$	$K^{-4} \cdot \begin{pmatrix} 0.1250 & 0.1250 \\ 0.1250 & 0.1875 \end{pmatrix}$	$\begin{pmatrix} 0.5774 & -0.8165 \\ 0 & 1.0000 \end{pmatrix}$	0.0980
$(\mathcal{P}_2, \Phi_2)$	$K^{-4} \cdot \begin{pmatrix} 0.3750 & 0.3750 \\ 0.3750 & 0.5625 \end{pmatrix}$	$\begin{pmatrix} 0.5774 & -0.8165 \\ 0 & 1.0000 \end{pmatrix}$	0.0980
$(\mathcal{P}_3, \Phi_3)$	$K^{-4} \cdot \begin{pmatrix} 0.7500 & 0.7500 \\ 0.7500 & 1.1250 \end{pmatrix}$	$\begin{pmatrix} 0.5774 & -0.8165 \\ 0 & 1.0000 \end{pmatrix}$	0.0980
$(\mathcal{P}_4, \Phi_4)$	$K^{-4} \cdot \begin{pmatrix} 1.2500 & 1.2500 \\ 1.2500 & 1.8750 \end{pmatrix}$	$\begin{pmatrix} 0.5774 & -0.8165 \\ 0 & 1.0000 \end{pmatrix}$	0.0980
$(\mathcal{P}_5, \Phi_5)$	$K^{-4} \cdot \begin{pmatrix} 0.7969 & 0.2813 \\ 0.2813 & 0.1875 \end{pmatrix}$	$\begin{pmatrix} 0.6860 & -0.7276 \\ 0 & 1.0000 \end{pmatrix}$	0.1203
$(\mathcal{P}_6, \Phi_6)$	$K^{-4} \cdot \begin{pmatrix} 2.2959 & 0.3828 \\ 0.3828 & 0.1875 \end{pmatrix}$	$\begin{pmatrix} 0.8121 & -0.5835 \\ 0 & 1.0000 \end{pmatrix}$	0.1508
$(\mathcal{P}_7, \Phi_7)$	$K^{-4} \cdot \begin{pmatrix} 0.9219 & 0.2813 \\ 0.2813 & 0.1875 \end{pmatrix}$	$\begin{pmatrix} 0.7365 & -0.6765 \\ 0 & 1.0000 \end{pmatrix}$	0.1318